ENHANCING SINGLE-CELL RNA-SEQ ANALYSIS THROUGH THE INTEGRATION OF UNACCOUNTED INTERGENIC REGIONS

Master Thesis

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Vilnius University

STUDENT NAME:	Juozapas Ivanauskas
STUDENT NUMBER:	2316457
SUPERVISOR:	Simonas Juzėnas, PhD
CONSULTANT:	Justina Žvirblytė, MSc

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Contents

1. LIST OF ABBREVIATIONS

cDNA	complementary DNA
CDS	coding sequence
CPM	counts per million
DGE	differential gene expression
GRN	gene regulatory network
NGS	next generation sequencing
ONT	Oxford Nanopore Technologies
ORF	open reading frame
PBMC	peripheral blood mononuclear cells
PCA	principal components analysis
RT	reverse transcription
scRNA-seq	single cell RNA sequencing
TF	transcription factor
t-SNE	t-distributed stochastic neighbor embedding
UMAP	uniform manifold approximation and projection
UMI	unique molecular identifier
UTR	untranslated region

2. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) is becoming an increasingly popular tool for analyzing cellular systems. This technology enables whole transcriptome sequencing of thousands of cells in a single experiment, generating a vast amount of data. The typical workflow for analyzing such data involves mapping reads to a known genome reference and constructing cell-gene matrices, i.e., matrices in which rows correspond to cells, columns correspond to genes, and each entry indicates the expression level of a gene in a specific cell. Such matrices are used in downstream analyses. However, there are some reads that map to the genome but remain unassigned to any known gene. As a result, such reads are not used in downstream analysis.

Since these unassigned reads can constitute a significant fraction of the total read count (often up to 30%), understanding their origin could help improve scRNA-seq data analysis.

There are two potential sources of unassigned reads: either they are sequencing artifacts, or they originate from real transcripts that remain unassigned due to issues related to transcriptomic references.

The two main challenges with transcriptomic references are:

1. Incomplete genomic annotations – even though great efforts have been made to annotate all genes, it is very likely that not all genes are annotated, and many remain to be found. Although potentially informative, reads from such unannotated regions are typically excluded from scRNA-seq analysis.

2. Complexity and overlaps in transcriptomic features – the genomes of humans are highly complex, with many overlapping genes. Reads from such overlapping regions are usually discarded from downstream analysis, due to the inability to assign unique gene per read.

This project focuses on unassigned reads in scRNA-seq datasets, aiming to uncover their origins and enhance scRNA-seq data analysis by incorporating biologically meaningful data that is typically disregarded.

3. AIM AND TASKS

Aim This project aims to determine whether including unassigned reads can improve scRNA-seq analysis.

Tasks

1. In scRNA-seq datasets, identify genomic regions containing unassigned reads, and classify them as either intersecting with genes or intergenic.
2. For intergenic regions, collect evidence to confirm whether they represent technical noise or putative unannotated genes.
3. For intersecting regions, attempt to resolve problems related to the transcriptomic reference that lead to reads being unassigned.

4. LITERATURE REVIEW

4.1 Introduction to single cell transcriptomics

Cells are the fundamental units of life, forming the basis of all living organisms. One of the major goals of biology is to understand cellular systems and the processes occurring within cells. Since the discovery of the DNA structure in 1953 and the development of the conceptual framework for genetic information transfer (Watson and Crick 1953), scientists have made significant efforts to sequence the genomes of various organisms. This led to the development of the first sequencing methods, such as Sanger sequencing in 1975 (Sanger, Nicklen, and Coulson 1977), which laid the foundation for next-generation sequencing (NGS) technologies in use today, such as the widely used Illumina platform (Heather and Chain 2016). Current sequencing methods allow us to obtain the complete genetic sequence of any organism. However, the genome alone cannot explain the full diversity of cells in multicellular organisms, as all somatic cells share the same genome but exhibit significant variation in shape, size, and function.

RNA sequencing (RNA-seq), on the other hand, enables the measurement of gene expression within cells, providing valuable insights into cellular processes. RNA-seq methods largely follow DNA sequencing protocols, with the addition of a step in which complementary DNA (cDNA) is synthesized from RNA (Heumos et al. 2023). The first RNA-seq methods were developed for bulk sequencing, where RNA from entire cell populations is sequenced, providing an average gene expression profile across the population. Although bulk RNA-seq has provided valuable insights into the dynamics of cellular processes (such as changes in disease states in response to therapeutics, detection of gene isoforms, gene fusions, and various other properties of target cells (Heumos et al. 2023)), this approach masks non-dominant processes and cell-to-cell variability through averaging. This limitation was addressed by the introduction of single-cell RNA sequencing methods, which allow for the generation of transcriptomic profiles from individual cells, providing high-resolution insights into cellular systems.

Current scRNA-seq methods enable the generation of transcriptomic profiles from thousands of cells at unprecedented resolution in a single experiment. These data can be used for constructing cellular atlases (Rozenblatt-Rosen et al. 2017), understanding disease mechanisms (Galen et al. 2019), exploring cell differentiation and developmental processes (Chu et al. 2016), among other applications.

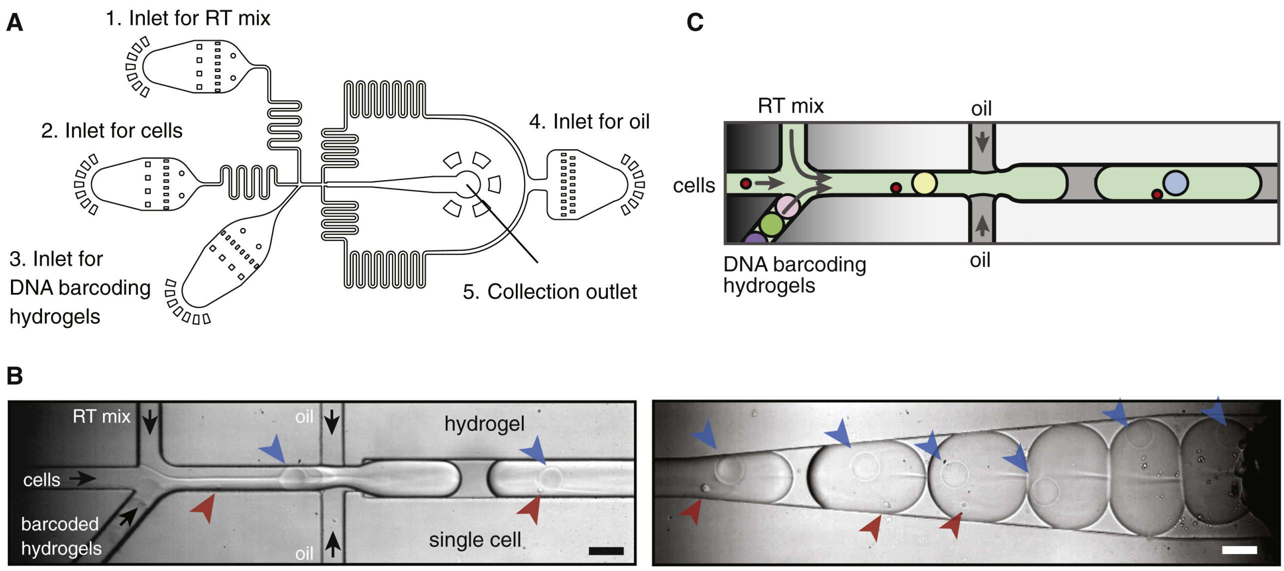


Figure 4.1: Example of a microfluidic device for droplet generation. (A) Schematic view of the device. It features four inlets (for reverse transcription mix, cell suspension, barcoding hydrogels, and oil) and one outlet for droplet collection. (B) Snapshots of droplet generation and collection. Blue arrows indicate hydrogels, red arrows indicate cells, and black arrows indicate flow direction. (C) Diagram of droplet generation. Figure taken from Klein et al. (2015).

4.2 scRNA-seq data generation and analysis

4.2.1 Overview of scRNA-seq protocols

The generation of scRNA-seq data is a complex, multi-step process that varies across different protocols. These protocols can be grouped based on several criteria, such as the type of RNA capture (e.g., 3' end, 5' end, or full-length) or the method of cell isolation (e.g., droplet-based methods such as inDrops (Klein et al. 2015), Drop-seq (Macosko et al. 2015), and Chromium by 10X Genomics (Zheng et al. 2017), or plate-based methods such as CEL-Seq2 (Hashimshony et al. 2016) and Smart-seq2 (Picelli et al. 2013)).

In this project, datasets generated using droplet-based 3' end sequencing methods were analyzed. Therefore, these methods will be the focus of the following literature review.

3' End Sequencing 3' end sequencing captures RNA molecules using primers complementary to poly-A tails. Polyadenylation is a post-transcriptional modification in which non-templated adenosines are added to the 3' end of RNA molecules. Although poly-A tail length can vary, it is present in nearly all mature RNAs (Brouze et al. 2022), enabling 3' end sequencing protocols to selectively capture transcripts in a high-throughput and cost-effective manner.

Common Steps in scRNA-seq Protocols Despite differences between scRNA-seq protocols, they share key steps: single-cell isolation, library preparation, and sequencing (Andrews and Hemberg 2018). In droplet-based methods, single cells are encapsulated in individual droplets containing hydrogel-attached barcoding primers and a lysis mix (an example of a droplet generation device is shown in Figure 4.1). Primers used in these protocols typically share a common structure, including:

5'- /5Acryd/ /iSpPC/ CGATGACG TAATACGACTCACTATAGGG ATACCACCATGG CTCTTTCCCTACACGACGCTCTTC
CGATCT [barcode1] AAGGCGTCACAAGCAATCACTC [barcode2] NNNNNN TTTTTTTTTTTTTTTTTTTV -3'

Figure 4.2: Primer design in the inDrops protocol. The image above shows schematic view, while below is given example with sequences. The structure of primers varies between the protocols, the main parts (UMI site, barcodes, poly-T's) are found in most of them. Here also present are promoter region for RNA polymerase (red), sequencing primer (blue), synthesis adapter (dark brown). Figure taken from Klein et al. (2015).

- **Cell barcodes:** Unique sequences that identify the cell from which a particular read originates (barcodes are needed as all droplets are pooled and sequenced together).
- **Unique molecular identifiers (UMIs):** Short sequences comprised of random nucleotides which are used to quantify the original number of RNA molecules, helping to eliminate amplification bias.
- **PCR handles:** Sequences that facilitate amplification.
- **Poly-T sequences**: In 3' end sequencing methods are used to selectively capture polyadenylated RNAs (Zhang et al. 2019).

An example of primer design is shown in Figure 4.2.

Once cells are encapsulated in droplets, lysis occurs, releasing RNA that is captured by barcoded primers. When the primers hybridize to the RNA molecules, reverse transcription (RT) takes place, generating barcoded cDNA. The emulsion is then broken, and the cDNA is amplified in bulk (typically using PCR) (Klein et al. 2015; Zheng et al. 2017). After amplification, the cDNA is processed to be compatible with the chosen sequencing platform. For short-read Illumina sequencing, this involves cDNA purification, fragmentation, and ligation of Illumina-compatible adapters.

4.2.2 scRNA-seq data analysis

Raw data processing The output from NGS machines is typically converted into FASTQ files, which contain the recorded sequences along with (depending on the protocol) barcode and UMI sequences, as well as quality scores.

The reads in these FASTQ files are then mapped to the reference genome. Although this step may seem straightforward, it is usually computationally intensive due to the size of the genome, the volume of reads, and the complexity of the data. For example, aligners for RNA data must be splicing-aware, and the input sequences may contain mismatches, insertions, and deletions caused by genomic variation and sequencing errors (Dobin et al. 2012).

Once mapped, reads are assigned to genes. This is done in a straightforward manner: if a read overlaps a gene in the transcriptomic annotation (typically provided in GTF format), it is assigned to that gene. However, complications arise when a read maps to multiple locations or to regions overlapping multiple genes — issues discussed further in Section 4.5.

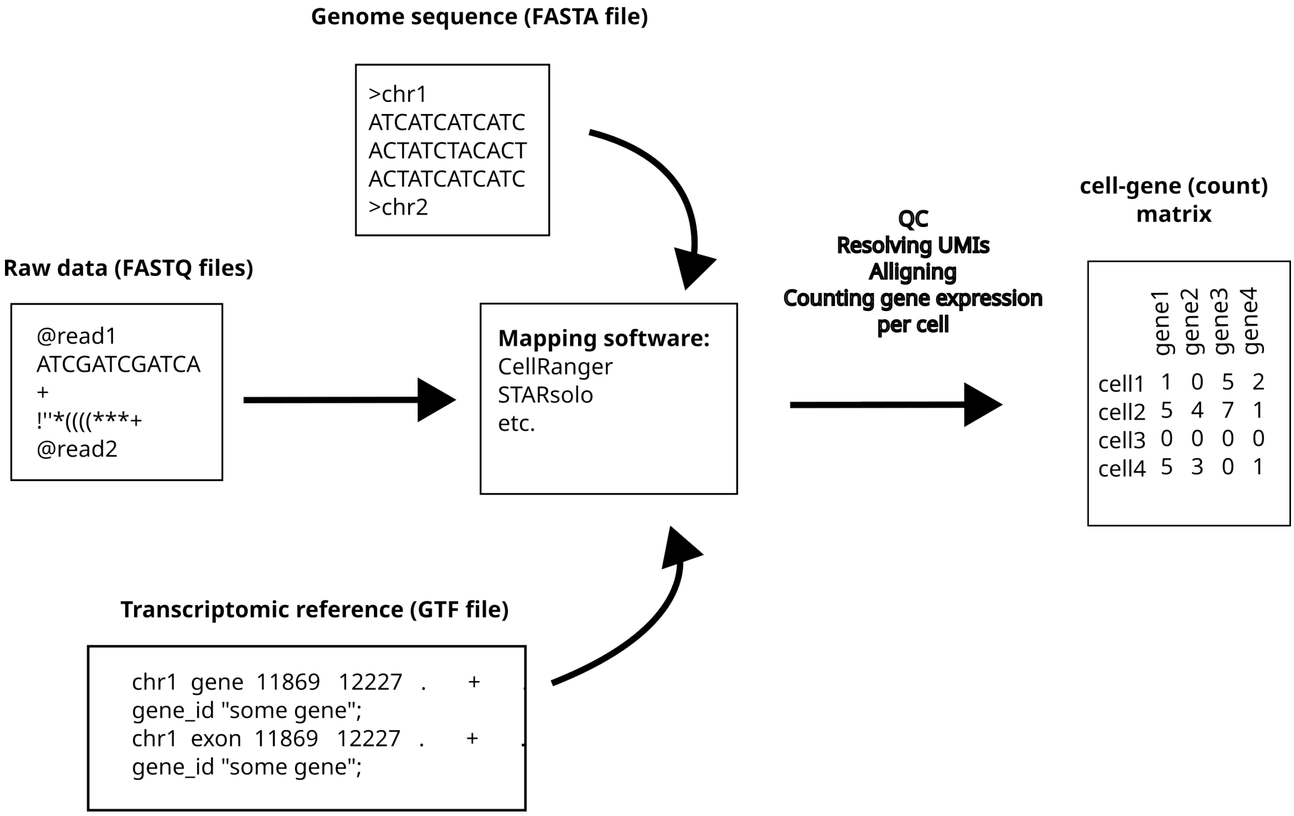


Figure 4.3: The process of mapping raw sequencing reads (FASTQ file) requires genome sequence (FASTA file) and transcriptomic reference (GTF file). The mapping software then alligns reads to the genome, and if there are genes present, assign them to the genes. Later the reads per cell are counted (using cell barcodes) and cell-gene matrix is constructed.

After gene assignment, duplicate reads are filtered out to estimate the original number of transcripts. This is achieved using UMIs: reads with the same UMI and cell barcode, mapping to the same gene, are considered duplicates. Additionally, reads with low base quality scores in UMI or cell barcode sequences are filtered out. Following this step, gene expression per barcode can be quantified, producing a cell–gene matrix (Heumos et al. 2023). See Figure 4.3 for a summary of the inputs and outputs in raw data processing.

These steps are typically performed using dedicated software packages such as STARsolo (Kaminow, Yunusov, and Dobin 2021), CellRanger (Zheng et al. 2017), or others. The exact pipeline may vary depending on several factors, such as the availability of a genome or transcriptome reference, and whether the protocol includes UMIs.

The typical output of this process is a cell–gene matrix, where rows represent cells, columns represent genes, and each entry reflects the number of captured RNA molecules from a given gene in a specific cell.

Cell-gene matrix processing The next steps in the analysis of the scRNA-seq data include following (see Figure 4.4 for graphical summary):

- **Quality control** The quality of individual cells (represented by all reads associated with a unique barcode) can be assessed based on several metrics, such as the proportion of mitochondrial gene expression (apoptotic cells tend to have higher levels of mitochondrial transcripts (Heumos et al.

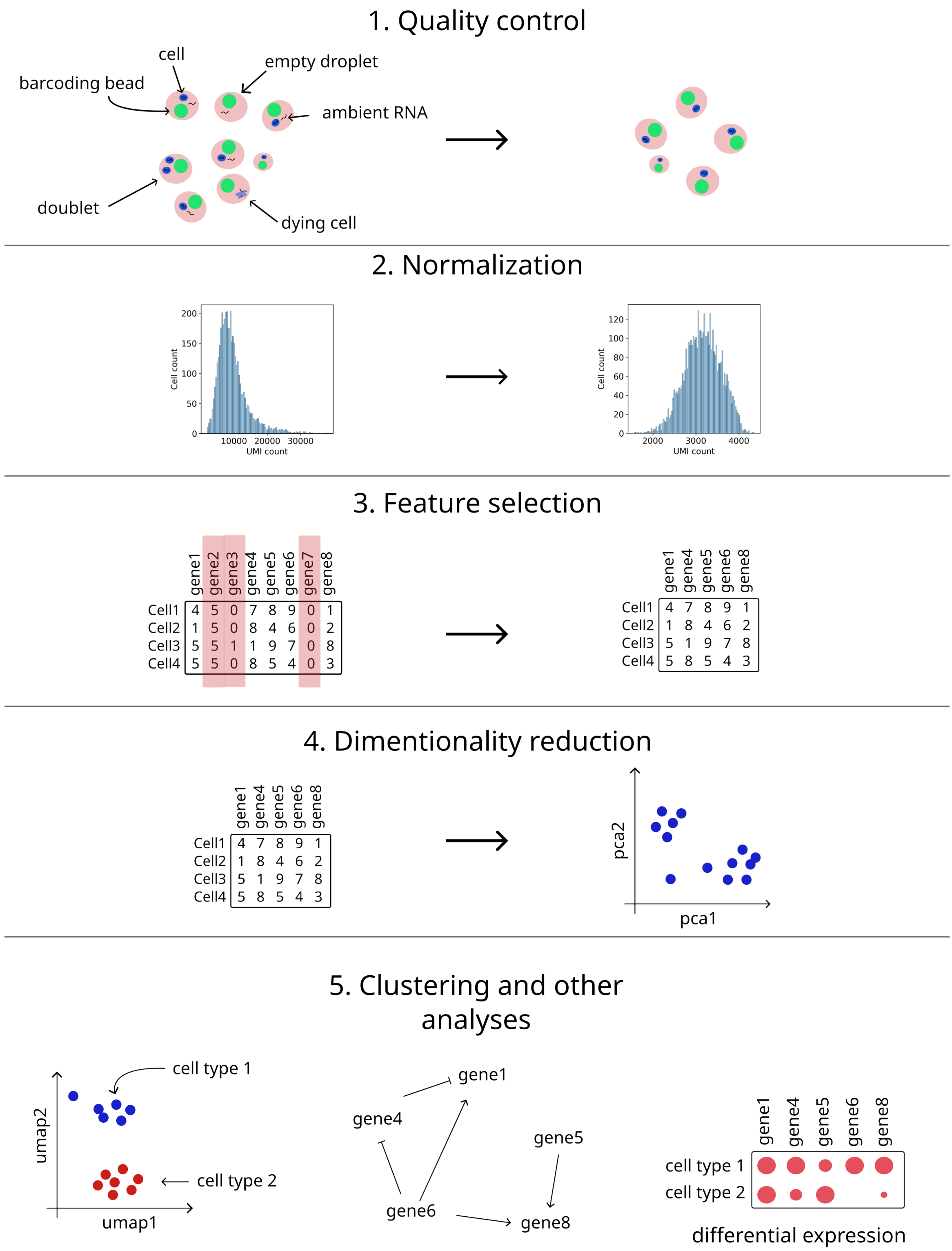


Figure 4.4: The typical data analysis pipeline. Firstly data is cleaned and normalized, then (to reduce dimensionality) non-informative genes are filtered out and dimensionality reduction methods are applied. Final steps depends on the focus of study, but often include cell clustering, annotation, differential gene expression analysis or gene regulatory network inference.

2023)) or the total number of captured transcripts. Occasionally, two cells may be encapsulated in a single droplet, resulting in a count matrix entry that reflects the combined gene expression of both cells. These entries, known as doublets, can be detected and removed using dedicated tools such as Scrublet (Wolock, Lopez, and Klein 2019) or scDblFinder (Germain et al. 2022). Another source of noise in scRNA-seq data is ambient RNA – cell-free RNA (e.g., released from dying cells) that disperses into the cell medium and may be encapsulated in droplets, contributing to background signal. Although the proportion of ambient RNA is typically low (around 2% in high-quality datasets (Young and Behjati 2020)), removing these transcripts from the count matrix can improve data quality. The background noise profile can be inferred from barcodes with very low transcript counts, as these likely represent empty droplets and thus consist entirely of ambient RNA. Several tools exist to model and correct for this background signal, including SoupX (Young and Behjati 2020), decontX (Yang et al. 2020), and CellBender (Fleming et al. 2023).

- **Normalization** The next step in the preprocessing pipeline is normalization, which aims to transform the data so that variation in gene expression levels becomes comparable across different genes and cells. This is important because standard statistical methods generally assume uniform variance in the data (Ahlmann-Eltze and Huber 2023). Normalization also helps correct for technical biases, such as differences in sequencing depth, which are especially relevant when combining data from multiple samples (Lingen, Suarez-Diez, and Saccenti 2024). Various normalization methods exist, based on different statistical principles, such as delta-method, residuals, latent gene expression modeling, or direct count adjustments (Ahlmann-Eltze and Huber 2023). General recommendations for normalization suggest comparing several methods, and if the results are similar, opting for the simpler method (Lingen, Suarez-Diez, and Saccenti 2024). Sophisticated methods do not necessarily show better results, and a recent benchmarking study by Ahlmann-Eltze and Huber (2023) has shown that simpler method (particularly the logarithm normalization, where each element y of the count matrix is transformed by formula $y_{transformed} = log(y + 1)$) performs as well or better than more advanced methods.

- **Filtering genes** Initially, count matrices include all genes present in the transcriptomic reference. While each gene is biologically meaningful, not all are useful for statistical analyses. For example, some genes may not be expressed in the sequenced cells or are expressed at very low levels (Heumos et al. 2023). It is therefore standard practice to filter out such genes — for instance, by excluding genes expressed in fewer than three cells. Additionally, some genes (commonly referred to as housekeeping genes) are expressed uniformly across most cells and thus provide limited discriminatory power for analyses such as cell clustering or cell type identification. To improve analytical focus and reduce computational complexity, it is often beneficial to retain only the most highly variable genes across cells. This reduces the dimensionality of the count matrix while preserving the most informative features for downstream statistical analyses.

- **Dimensionality reduction** Even after filtering and retaining only highly variable genes, several thousand features usually remain. Visualizing and interpreting such high-dimensional data is challenging, making dimensionality reduction a crucial step in the analysis pipeline. The goal of dimensionality reduction is to project the data into a lower-dimensional space while preserving as much relevant structure and variability as possible. Various methods exist for this purpose, each based on different mathematical principles. Among the most widely used are principal component analysis (PCA), t-distributed stochastic neighbor embedding (t-SNE) (Hinton and Roweis 2002), and uniform manifold approximation and projection (UMAP) (McInnes, Healy, and Melville 2018). Benchmarking studies provide mixed results regarding their performance. For instance, Xiang et al. (2021) reported that t-SNE shows best perfomance overall, while UMAP offers greater stability. In contrast, Koch et al. (2021) found that less commonly used methods such as latent Dirichlet allocation (LDA) and PHATE outperformed others in terms of clustering accuracy, structure preservation, and computational efficiency. Similarly, S. Sun et al. (2019) proposed that the choice of dimensionality reduction method should depend on the specific downstream analysis, and in their evaluation, UMAP and t-SNE were not among the top recommendations. Despite this, UMAP remains the de facto standard in many workflows (typically applied after an initial PCA step) but exploring alternative methods may yield better results depending on the context.

- **Clustering and other analyses** One of the most common tasks in scRNA-seq data analysis is identifying and classifying cell populations (Andrews and Hemberg 2018). This involves assigning cells to groups (clusters) such that cells within the same cluster are similar to each other and distinct from those in other clusters. A wide range of clustering algorithms is available, including k-means, hierarchical, and consensus clustering methods (Peng et al. 2020). Benchmarking studies indicate that no single algorithm consistently performs best across all scenarios (Peng et al. 2020).

  Clustering is typically followed by cell type annotation (i.e. assigning biological identities to cell clusters) either by identifying known marker genes or using automated tools such as CellTypist (Domínguez Conde et al. 2022). Subsequent analysis steps depend on the study's objectives and may include differential gene expression (DGE) analysis, exploration of cellular dynamics (e.g., RNA velocity, pseudotime analysis), inference of gene regulatory networks (GRNs), and more.

The described analysis pipeline enables insights into multicellular systems. However, all downstream analyses critically depend on the accuracy of the initial processing steps, especially read mapping, which in turn relies on the quality of the transcriptomic reference. In the following sections, I will shortly introduce genome biology and further expand on transcriptomic references.

4.3 Genome biology

The RNA molecules captured in scRNA-seq experiments reflect the transcriptional output of the genome. RNA is considered either a final or intermediate gene product, depending on whether the gene encodes a functional RNA or a protein. Genes are typically defined as genomic loci transcribed into functional RNA molecules.

In eukaryotes, primary transcripts often undergo *splicing*, a process in which segments called *introns* are removed and *exons* are joined to form the mature RNA (see Figure 4.5). While introns are not retained in the final transcript, they can contribute to gene regulation, affecting the rate of transcription, nuclear export, and transcript stability (Shaul 2017). Importantly, the classification of

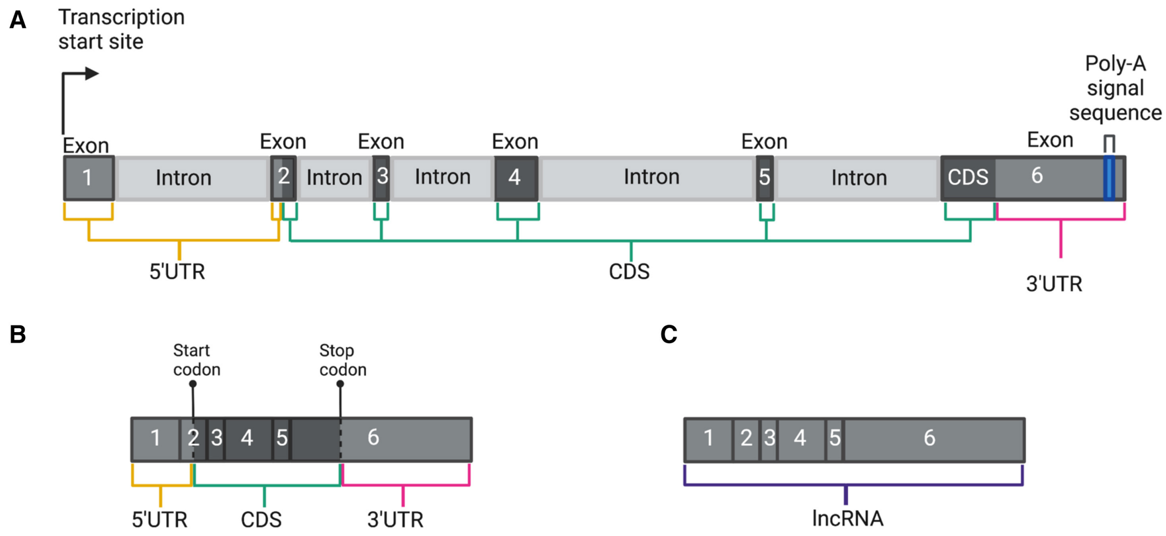


Figure 4.5: Schematics showing an example of gene structure. A) A genomic region of a protein-coding gene. It consists of 6 exons: the 3rd, 4th, and 5th are entirely CDS; the 2nd and 6th contain both CDS and UTR; and the 1st is entirely non-coding (5' UTR). B) The mature RNA of protein-coding transcript represented in (A). C) The mature long non-coding RNA consists solely of non-coding exons. Figure taken from Aspden, Wallace, and Whiffin (2023).

exons and introns is transcript-specific: due to alternative splicing, a genomic region may be exonic in one transcript and intronic in another (Aspden, Wallace, and Whiffin 2023).

Exons that encode the amino acid sequence of proteins are referred to as *coding sequences* (CDS), while non-coding exons may contribute regulatory elements. A typical protein-coding gene includes *untranslated regions* (UTRs) at both the 5' and 3' ends. Although UTRs do not encode proteins, they are functionally important, influencing mRNA localization, stability, and translation (Mayr 2018; Hinnebusch, Ivanov, and Sonenberg 2016).

Transcription Transcription is a tightly regulated process, as different cell types require the expression of distinct sets of genes. This regulation is largely governed by non-coding DNA regions, particularly promoters and enhancers.

Promoters are DNA elements located near the transcription start sites of genes, serving as binding platforms for RNA polymerase and other components of the transcriptional machinery. Their activity depends on the presence and availability of specific transcription factors (TFs) in the nucleus (Cramer 2019).

Enhancers are another class of regulatory elements that can influence transcription from a distance, often located hundreds of kilobases or even over a megabase away from their target genes (Furlong and Levine 2018). They function by recruiting transcription factors and coactivators, facilitating chromatin looping or condensate formation to bring them into physical proximity with promoters, thereby increasing transcriptional activity in a context-dependent manner (Kittle and Levo 2025).

TFs are proteins that bind specific DNA sequences to modulate transcriptional output. Some directly recruit RNA polymerase, while others function by recruiting cofactors that alter chromatin accessibility or facilitate specific phases of the transcription cycle (Lambert et al. 2018).

Transcription of many genes does not occur as a continuous, uniform process but rather in discrete bursts. This phenomenon, known as transcriptional bursting, involves episodic periods of transcriptional activity interspersed with silent intervals. Bursting is a key contributor to cell-to-cell variability in gene expression, or transcriptional noise, especially in genes with low expression levels. The frequency and size of these bursts can be modulated by promoter architecture, enhancer activity, chromatin state, and TF binding dynamics (Hebenstreit and Karmakar 2024).

In addition to dynamic regulation, the transcription process is inherently prone to errors. RNA polymerase can misincorporate nucleotides, leading to transient errors in RNA sequences (Gordon et al. 2015). Although these transcription errors are not propagated across cell generations like DNA mutations, they can affect the function of transcripts and proteins, particularly in highly expressed or functionally sensitive genes. Sources of transcription errors include polymerase infidelity, sequence context, and stress conditions, among others. Some transcription errors may even lead to the production of neoepitopes or trigger RNA surveillance pathways, highlighting their potential biological consequences (Y. Sun and Vermulst 2025).

Chromatin For RNA polymerases to access DNA, the structural organization of the genome is critical. This organization is governed by chromatin, a dynamic complex of DNA and histone proteins that compacts and regulates the eukaryotic genome within the nucleus. Its fundamental unit, the nucleosome, consists of approximately 147 base pairs of DNA wrapped around a histone octamer, allowing both efficient packaging and regulatory control (Alberts et al. 2022).

Chromatin exists along a continuum of conformational states, ranging from tightly packed heterochromatin, which is generally transcriptionally silent, to more relaxed euchromatin, which is associated with active transcription. Accessible chromatin refers to regions where DNA is less tightly associated with histones, allowing interaction with transcription factors and other regulatory proteins. Although such accessible regions comprise only 2–3% of the genome, they harbor over 90% of known transcription factor binding sites (Klemm, Shipony, and Greenleaf 2019).

Chromatin also contributes to the higher-order three-dimensional (3D) organization of the genome. This structure is not random but is hierarchically organized across multiple scales. For instance, individual chromosomes occupy distinct territories within the nucleus, with their positioning correlating with gene activity and cell-type-specific functions. Within these territories, genomic regions are further organized into topologically associating domains, which facilitate or restrict interactions between enhancers, promoters, and other regulatory elements. Such spatial organization plays a crucial role in gene regulation, including long-range interactions between loci on the same or even different chromosomes (Elimelech and Birnbaum 2020).

Genome-wide assays such as ATAC-seq and DNase-seq are widely used to map chromatin accessibility, providing insight into regulatory landscapes and cell-type-specific transcriptional programs (Minnoye et al. 2021).

Gene classification RNA transcripts vary in function and biological role. A foundational classification distinguishes genes as either protein-coding or non-coding. While this binary distinction remains common, it is increasingly seen as oversimplified. Protein-coding genes can yield non-coding isoforms, and some non-coding genes contain small open reading frames (sORFs) that are translated into functional peptides (Poliseno, Lanza, and Pandolfi 2024).

Protein coding genes Protein-coding genes are the most extensively studied and typically share common structural features, including 5' and 3' UTRs and an open reading frame (ORF), which is translated into a polypeptide. Their expression and processing follow the classical central dogma of molecular biology: DNA is transcribed to mRNA, which later is translated to protein. Current annotations converge on an estimate of approximately 20,000 protein-coding genes in the human genome (Amaral et al. 2023).

Non-coding genes As mentioned earlier, not all transcribed RNAs are translated into proteins. In fact, a substantial portion of the genome is transcribed into non-coding RNAs, which include a wide range of functional RNA types. Among these, long non-coding RNAs (lncRNAs) have emerged as a major output of complex eukaryotic genomes (Mattick et al. 2023). These RNAs are typically longer than 200 nucleotides and are not translated into proteins, yet they participate in essential biological processes.

Non-coding RNAs can serve both housekeeping and regulatory roles. Housekeeping ncRNAs include rRNAs, tRNAs, and small nuclear RNAs, while regulatory ncRNAs are involved in gene expression regulation, chromatin remodeling, RNA splicing, translation, and even phase separation in the nucleus. Long non-coding RNAs (lncRNAs) are often transcribed from genomic regions located between protein-coding genes, but they can also arise from introns, antisense strands of other genes, pseudogenes, and enhancers (Poliseno, Lanza, and Pandolfi 2024).

These RNAs tend to be expressed in a cell type-specific and developmentally regulated manner, contributing to lineage specification, tissue identity, and responses to environmental cues. Despite their relatively low abundance and rapid sequence evolution, lncRNAs often exhibit conserved structure-function features, such as modular domains and repeat elements that mediate interactions with chromatin, RNA, and proteins. Importantly, they are frequently associated with chromatin-modifying complexes and are increasingly recognized as integral to the spatial and temporal organization of gene regulation during development (Mattick et al. 2023).

Given their diversity and specificity, the biological importance of non-coding RNAs, especially lncRNAs, is increasingly appreciated, and many are now considered as genes with crucial regulatory roles in development, physiology, and disease.

Other genome components The functional elements described so far account for only a fraction of the genome. A large portion of the genome is composed of repetitive sequences, which are not unique. In humans, more than 50% of the genome consists of various classes of repeats (Liehr 2021). These were historically referred to as 'junk', 'selfish', or 'parasitic' DNA. However, subsequent research has revealed that many repetitive elements have structural, regulatory, or evolutionary functions. Some contribute to genome stability by providing redundancy or scaffolding, while others influence gene expression or chromatin structure (Shapiro and Sternberg 2005).

A substantial fraction of these repeats originate from transposable elements (transposons) — mobile DNA sequences capable of moving or copying themselves within the genome. Once thought to be purely parasitic, transposons are now recognized for their generative role in genome evolution, and are increasingly viewed as integral components of genome architecture (Fedoroff 2012).

Pseudogenes are believed to arise, in part, from transposon-mediated duplication or retrotransposition events. They are genomic regions that closely resemble functional genes but contain disabling mutations or lack the necessary regulatory elements for transcription or translation (Qian et al. 2022). Initially considered evolutionary relics, pseudogenes are now known to exhibit diverse biological roles. Some have been implicated in the regulation of gene expression, particularly through RNA-mediated mechanisms (Tutar 2012). Moreover, it is estimated that up to 19% of human pseudogenes may be translated, possibly contributing to the emergence of new proteins (Qian et al. 2022).

Other repetitive structures include centromeres and telomeres, which are composed of tandem repeats and play essential roles in chromosome segregation and end protection, respectively, during DNA replication and cell division (Alberts et al. 2022).

4.4 Genome annotation

The genome annotation process typically refers to the identification and mapping of genes within a given genome sequence (Guigó 2023). While the definition itself is straightforward, the process is highly complex. This is evident from the fact that, even more than 20 years after the first human genome assembly, human genome annotations are continuously updated with new transcripts and are expected to evolve further (Mudge et al. 2024). Figure 4.6 illustrates the changes in GENCODE annotation over time.

It is important to note that many medical and scientific research efforts rely on an accurate human gene list. Examples include genome-wide association studies (GWAS), which attempt to link genomic variants to nearby genes; RNA-seq analysis; and exome sequencing projects that use capture kits targeting most known exons (Pertea et al. 2018).

Currently, the two most widely used genome annotations are GENCODE (curated by the GENCODE consortium at EMBL–EBI and Ensembl) (Mudge et al. 2024) and RefSeq (O'Leary et al. 2015) (maintained by NCBI). Despite their status as mature genome references, they report different numbers of genes. For instance, RefSeq includes 20,078 protein-coding genes (NCBI 2025), whereas GENCODE contains 19,868 (Ensembl 2025). This discrepancy highlights the ongoing challenge of accurately annotating genomes.

Genome annotation process RNA sequencing (RNA-seq) is the primary tool used in genome annotation. Full-length RNA sequencing allows for the capture of RNA molecules, which, when aligned to a reference genome, help construct the genome annotation of a given species (Salzberg 2019). However, RNA-seq has limitations, the most significant being its inability to capture all RNA molecules. This poses particular challenges for detecting rare transcripts, which may either be treated as noise or not captured at all (Salzberg 2019). Therefore, bioinformatics tools are necessary to complement RNA-seq data (Guigó 2023), and most modern genome annotation pipelines integrate computational methods alongside sequencing data.

Computational genome annotation methods can be broadly categorized into two types: comparative annotation, which leverages the fact that protein-coding sequences tend to be more evolutionarily conserved, and *ab initio* annotation, which uses known sequence biases to predict genes (Guigó 2023). Despite significant advancements, genome annotation remains imperfect.

While the number of protein-coding genes is reaching a consensus (major databases report around 19,000 to 20,000 protein-coding genes) the number of non-coding genes is expected to increase in the future (Amaral et al. 2023). This is largely due to the structured nature of protein-coding genes (e.g., open reading frames, codon biases that skew nucleotide distributions), which makes them easier to detect using computational tools (Guigó 2023). Additionally, protein-coding genes have historically received greater attention because of their direct links to phenotype, leading to more experimental validation.

Current challenges in human genome annotation Despite extensive efforts, a fully comprehensive human genome annotation has not yet been achieved. Several factors contribute to this challenge.

Firstly, the complexity of the human genome itself presents significant obstacles. Compared to the total genome length, the number of genes is relatively low, and they are organized as a series of separated exonic regions (Salzberg 2019). Additionally, bulk RNA sequencing often has relatively low sequencing depth, making it difficult to detect rare transcripts or those expressed in a cell-type-specific manner (Guigó 2023).

In principle, long-read single-cell RNA sequencing could alleviate some of these limitations, as long reads are more easily mappable. However, this approach is still constrained by high error rates and low sequencing depth (Guigó 2023). In contrast, short-read scRNA-seq generally offers higher throughput (Heumos et al. 2023), facilitating the detection of rare transcripts. Yet, it fails to capture full-length transcript structures, limiting its utility to supporting roles, such as validating predicted genes or indicating transcriptional activity at specific genomic loci.

Beyond technical limitations, there are ontological challenges, such as defining what constitutes a gene. While genes are often regarded as well-defined, discrete entities, the reality is more complex. Coding and non-coding transcripts frequently overlap in intricate arrangements with unclear boundaries, suggesting that transcripts may not be discrete, countable units but instead form a transcriptional continuum (Salzberg 2019). Additionally, the classification systems used in genome annotations may be partially artificial. For example, some pseudogenes, generally considered non-functional copies of functional genes, are transcribed and may have biological functions (Pei et al. 2012). Similarly, the distinction between protein-coding and non-coding genes is debated, as many protein-coding loci generate both coding and non-coding transcripts, and numerous long non-coding RNAs (lncRNAs) contain potentially coding open reading frames (ORFs) (Salzberg 2019).

Future perspectives of annotating genomes The size of eukaryotic genomes necessitates the use of automated methods for genome annotation. The accuracy of such methods depends directly on RNA capture technologies. If highly accurate and sensitive methods are developed, high-quality genome annotations can be expected (Salzberg 2019). However, at present, manual curation of human genome annotations remains essential due to the imperfections of both sequencing technologies and computational tools. Given the critical role of accurate gene annotation in both medical and scientific applications, continued improvements in annotation methods remain a priority.

4.5 Transcriptomic references for scRNA-seq

A well-annotated genome does not eliminate all challenges during read mapping. In simple terms, mapping algorithms determine the genomic location to which a sequence aligns, and if an annotated gene is present at that location, the read is assigned to that gene. However, there are cases where

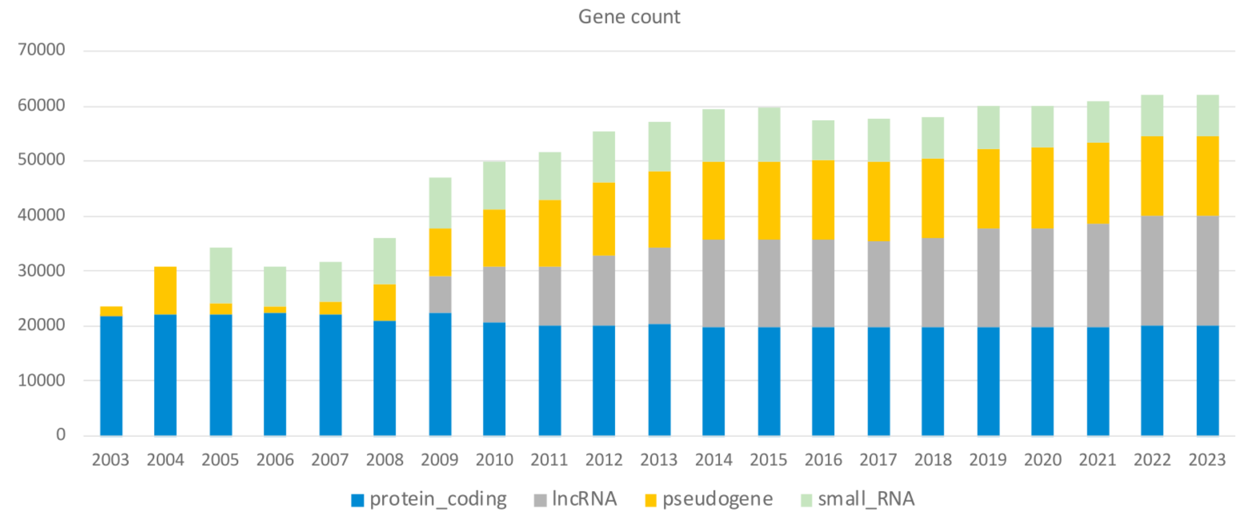


Figure 4.6: Changes in the number of annotated genes in GENCODE over time. Figure taken from paper by Guigó (2023)

assigning a read to a gene is not straightforward, such as when a read maps to multiple locations or to a unique location where overlapping genes are annotated.

Multimappers There are two possible reasons for reads assigned to multiple genes: either they are mapped to multiple genome locations, or they are mapped to the genomic location containing overlapping features (genes). Such reads (commonly referred to as multimappers) are typically excluded from analysis (Almeida da Paz, Warger, and Taher 2024). However, as shown by Almeida da Paz, Warger, and Taher (2024), this exclusion can introduce biases that affect downstream analyses. For instance, highly similar gene families may be underrepresented; in the case of the ubiquitin B family, only 3% of reads mapping to these loci are uniquely assigned. Although various methods have been proposed to handle multimappers (Deschamps-Francoeur, Simoneau, and Scott 2020), no fully satisfactory solution has yet been established (Almeida da Paz, Warger, and Taher 2024).

Multimapper resolution is primarily addressed through computational strategies. However, the issue can also be mitigated by refining the transcriptomic reference itself. For example, in the case of CellRanger (the mapping software from 10x Genomics) the GENCODE reference is filtered based on gene types (e.g., retaining protein-coding genes while filtering out pseudogenes) (10x Genomics 2025). While this filtering improves the number of reads included in downstream analyses, there is still room for improvement.

Pool et al. (2023) proposed three key steps to enhance transcriptomic references:

1. Including reads mapped to intronic sequences in the analysis.
2. Extending the 3' ends of certain genes.
3. Resolving overlaps between specific genes.

The first suggestion is not new in scRNA-seq research. Concepts such as RNA velocity, which rely on the ratio of spliced to unspliced RNA (La Manno et al. 2018), demonstrate that including intronic reads can provide valuable information. Moreover, most mapping tools (e.g., STARsolo, CellRanger) offer options to align reads either to exonic regions only or to entire genes. The second suggestion is based on the observation that scRNA-seq data often exhibits peaks of reads just beyond the 3' ends of genes. While the exact biological reasons remain unclear — possibly due to imprecise annotations — it is reasonable to associate these reads with the nearest genes. The third suggestion addresses gene overlaps. Reads originating from overlapping regions are often unassigned to any gene, yet in some cases, they are more likely to come from one gene rather than another. Overlapping gene resolution seeks to correct this by modifying the transcriptomic reference, either by shortening or removing certain genes.

Although Pool et al. (2023) proposed a tool to implement these strategies, it has limitations. Some of its aspects remain debatable (e.g., threshold choices), others seem unnecessary (such as handling exon-intron distinctions when most alignment tools already provide this option), and the process still requires significant manual effort. Thus, there is still a need for a more comprehensive tool for enhancing transcriptomic references.

Reads mapped to the intergenic regions Some reads map to the genome but fall outside of annotated genes. While such reads are often considered noise in scRNA-seq data, they may also represent biologically meaningful signals. As will be demonstrated in the results section, scRNA-seq datasets do contain these reads, and they are not merely technical artifacts.

5. METHODS

5.1 Data Acquisition

scRNAseq datasets We analyzed datasets from four different human tissues: brain, blood, lung, and eye. All samples, except for two Peripheral Blood Mononuclear Cell (PBMC) datasets, were generated using the 10x Genomics v3.1 protocol. The two exceptions were prepared using the inDrops2 protocol (Juzenas et al. 2025). All protocols capture 3' ends of RNA molecules and use short-read sequencing. The datasets are publicly available, with sources and additional details provided in Table 5.1.

sample name	tissue	cell count	donors	protocol	source
PBMC_10x	blood (PBMC)	5000	1	10x v3.1	10x genomics
PBMC_10x_2	blood (PBMC)	10000	1	10x v3.1	10x genomics
PBMC_10x_3	blood (PBMC)	10000	1	10x v3.1	10x genomics
PBMC_indrops	blood (PBMC)	2000	1	inDrops2	Juzenas et al. 2025
PBMC_indrops_2	blood (PBMC)	9000	1	inDrops2	Juzenas et al. 2025
brain	brain	6000	1	10x v3.1	Siletti et al. 2023
brain_2	brain	7000	1	10x v3.1	Siletti et al. 2023
eye	retina	10000	6	10x v3.1	Menon et al. 2019
eye_2	peripheral retina	2500	1	10x v3.1	Voigt et al. 2019
eye_3	peripheral retina	2500	1	10x v3.1	Voigt et al. 2019
lung_2	lung	5000	1	10x v3.1	Mould et al. 2021
lung_5	lung	5000	1	10x v3.1	Mould et al. 2021
lung_7	lung	4500	1	10x v3.1	Mould et al. 2021
lung_8	lung	5000	1	10x v3.1	Mould et al. 2021

Table 5.1: Datasets summary.

Genome and Transcriptomic references The human genome assembly GRCh38 was used in this project, downloaded from the Ensembl website.

Four transcriptomic references were analyzed (see Table 5.2 for details). To ensure compatibility with the genome FASTA file, chromosome name prefixes ('chr') were removed from references that included them, as the genome file does not use these prefixes. This modification was performed using basic Linux command-line tools. Additionally, for the NCBI reference, chromosome names were converted to Ensembl-style notation using a custom Python script (e.g., 1, 2, 3 instead of NC_000001.11, NC_000002.12, NC_000003.12, etc.).

For references that do not have 'gene' entries, such entries were added (entries that span all the components of a particular gene).

reference name	source	version
10x	10x website	2024-A
GENCODE	Gencode website	47
RefSec (NCBI)	NCBI website	GCF_000001405.40
LNCipedia	LNCipedia website	5.2

Table 5.2: References used in this project.

Gene Prediction Tracks and Conservation Scores Gene predictions (tracks 'AUGUSTUS', 'Geneid genes', 'Gescan genes', 'SGP genes', 'SIB genes') and genome conservation scores ('phastCons100way' track) were downloaded from UCSC genome browser.

ATAC data The open chromatin regions (bed format) in PBMCs was downloaded from 10x website.

Oxford Nanopore Technologies (ONT) data PBMC ONT data (full-length cDNA) were downloaded from 10x genomics website.

5.2 Computational Tools and Environment

All analyses were performed on a high-performance computing (HPC) cluster running a Linux environment. Software packages and tools were managed using Conda. The exact Conda environment specifications (YAML file) can be found on GitHub repository. Besides tools managed by conda, also basic command line tools were used (e.g. *awk, wc, grep, sed, uniq, sort* and similar). The specific functions and parameters used are described in the following sections, where the general analysis pipeline will be explained.

5.3 Data processing pipeline

The pipeline was implemented using GNU Make, and the full scripts are available on GitHub. Here, only a general description of the workflow and tools used is provided.

The overall structure of the pipeline, which was applied both for extracting intergenic regions and enhancing the transcriptomic reference, is shown in Figure 5.1, with a step-by-step description provided below:

1. Map reads using initial transcriptomic reference.
2. Extract unassigned and uniquely mapped reads.
3. Separate reads into two categories: intersecting with genes and intergenic.
   (a) For intersecting:
      i. Resolve cases of overlapping genes that also overlap with unassigned reads.

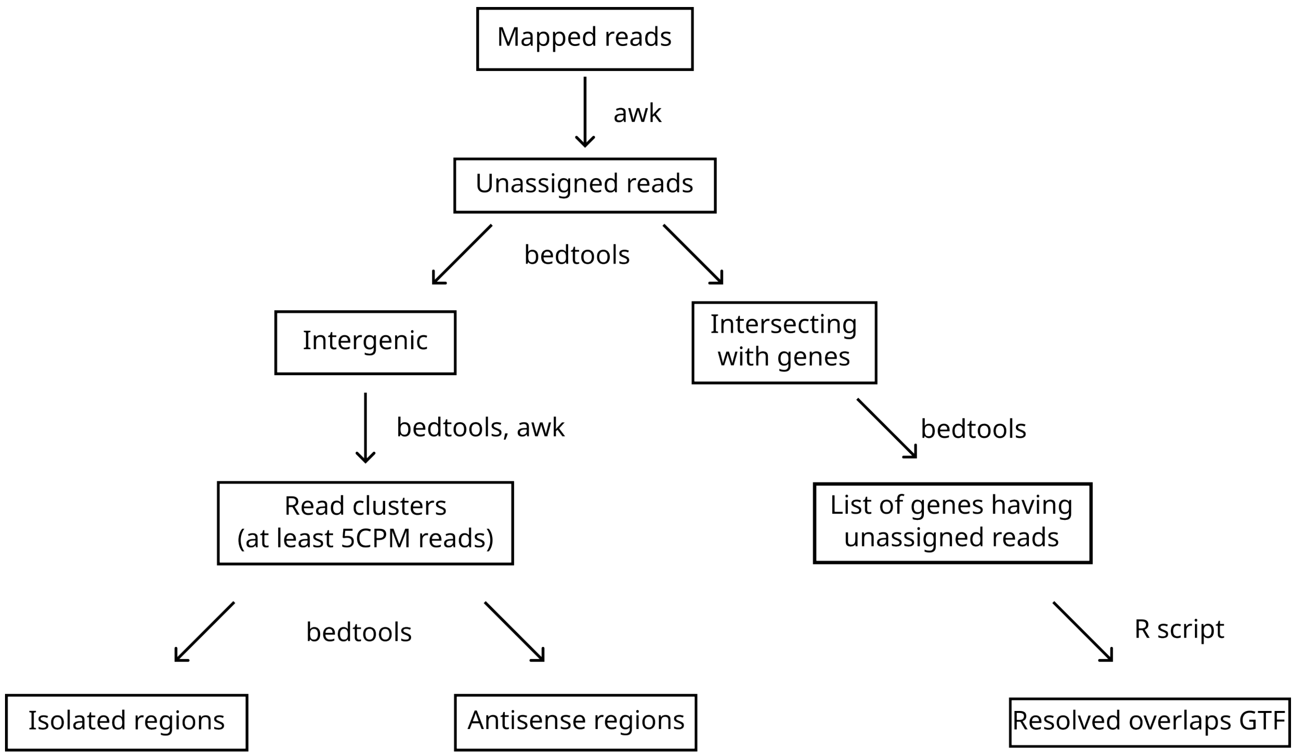


Figure 5.1: Schematic overview of the pipeline. This diagram illustrates the general logic and main tools used in the data processing workflow. The actual pipeline is more complex — for example, the use of multiple transcriptomic references required looping over some of the steps.

ii. From the second reference onward, add genes to the original transcriptomic reference if they:
    - overlap with unassigned reads, and
    - do not overlap with entries in the original reference.

(b) For intergenic:
  i. Cluster the reads.
  ii. Filter out small clusters (custom threshold).
  iii. For the first reference only: filter-out AT-rich reads.
  iv. For the remaining reads, repeat the process with the next reference.
  v. For the final reference only: assign clusters that start immediately downstream of 3' ends of known genes to those genes (by extending gene annotations in the reference).

4. Construct the final transcriptomic reference and remap the original reads to it.

5. Generate a list of large intergenic clusters (custom threshold).

Mapping and filtering BAM files Reads were mapped using STARsolo (Kaminow, Yunusov, and Dobin 2021) from both FASTQ and BAM input files. The same parameters were used across all samples (see the GitHub repository), except for those related to barcode structure — differences included the number of barcodes, lengths of barcode/UMI sequences, and their positions in the primers. Unassigned reads were filtered using *awk* by selecting those that:

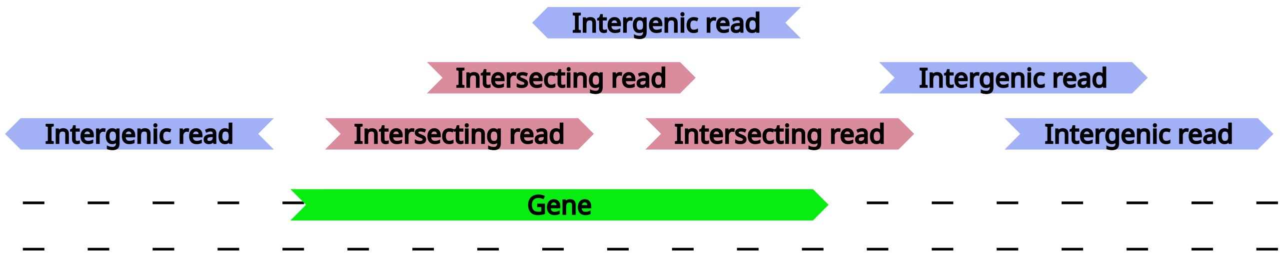


Figure 5.2: Classification of unassigned reads as 'intergenic' or 'intersecting'. Reads are classified as intersecting if they overlap with known genes and are on the same strand. Reads that do not meet these criteria are classified as intergenic.

1. had valid barcodes (i.e., barcode length matched the expected value)
2. were not assigned to any gene (tagged with `GN:Z:-` in the BAM file)
3. were uniquely mapped (tagged with `NH:i:1` in the BAM file).

Classifying reads as 'intergenic' or 'intersecting' and read clustering Unassigned reads were classified as either 'intersecting' or 'intergenic' based on their overlap with the reference genome, using the *bedtools intersect* command. Reads that intersected with genes were categorized as 'intersecting', while those that did not intersect were classified as 'intergenic'. Intersections were assessed in a strand-specific manner using the `-s` flag in *bedtools*, meaning that regions on different strands were treated as non-intersecting, even if they shared overlapping genomic coordinates (see Figure 5.2). Intergenic reads were then clustered using *bedtools merge*, again considering strand specificity.

Manipulating transcriptomic references The process of resolving overlapping genes and constructing the enhanced transcriptomic reference was carried out using a custom R scripts, specifically utilizing the *rtracklayer* library.

5.4 Intergenic regions

Extracting intergenic regions The intergenic reads were extracted as described in the previous section (i.e., unassigned reads that do not overlap with any reference used) and clustered using the *bedtools merge* command. Clusters were filtered based on their size, requiring at least 5 reads per million (CPM), where CPM represents the normalized read count per million total primary reads in the dataset (i.e., $1CPM = 1 \times sample\_size/1000000$). This filtering was performed using *awk*.

Cluster locations were adjusted using *deeptools* (for coverage calculation) and a custom Python script to prevent long introns from skewing the width of the intergenic clusters. To accomplish this, for each intergenic region, the maximum coverage location was identified and extended to include neighboring regions with at least half of the maximum coverage.

The lists from all samples were then merged using the *bedtools merge* function. This combined list was further filtered to retain only regions detected in at least all samples from one tissue group (e.g., *eye*, *lung*, *brain*, *PBMC_10x*, or *PBMC_indrops*). This filtering allows for the inclusion of tissue-specific regions while removing those that are not persistent across the same sample type. Additionally, for each entry, we determined whether it overlaps with predicted genes from the UCSC Gene Prediction

Figure 5.3: Classification of intergenic regions as antisense or isolated.

Archive (using *bedtools merge*) and calculated the distances to the closest genes (using *bedtools closest*). Intergenic regions located on the opposite strand of known genes were classified as *antisense*, while those that were not were classified as *isolated* (see Figure 5.3 for illustration).

This filtered combined list was then converted into GTF format, and unassigned reads from each sample were mapped using this combined intergenic annotation.

Analysis of cell-gene matrices The matrices produced by STARsolo were filtered to retain cells with a sufficient number of reads (thresholds selected manually) and genes expressed in at least three cells. Additionally, cells were filtered based on mitochondrial gene content, allowing up to 10% of mitochondrial reads per cell. Doublets were identified and removed using *scrublet*.

After filtering, the matrices were normalized using the *normalize_total* function from the *scanpy* package and log-transformed (i.e., each entry x was transformed as $x = \log(x+1)$ using the *log1p* function from *scanpy*).

Subsequently, only highly variable genes were retained (with parameters: `min_mean=0.0125`, `max_mean=3`, `min_disp=0.5`). For clustering, principal components (PCs) were computed, and an optimal number was selected manually based on the elbow rule. In *PBMC_10x* samples, cells were annotated automatically using CellTypist. For other samples, annotation was omitted, as it was not the main focus of the project. Visualizations were generated using UMAP embeddings, also implemented via *scanpy*.

All scripts and additional descriptions are available in the GitHub repository.

Correlations Spearman's rank correlation between intergenic regions and nearby genes was calculated using a custom Python script. The formula for Spearman's correlation is:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$$

where d_i is the difference between the ranks of the corresponding values, and n is the number of observations.

Nanopore data ONT data were aligned to the genome using the *minimap2* algorithm with the `-ax splice` option and possible transcript structures were inferred using *stringtie* transcript assembler tool with default parameters.

5.5 Enhancing transcriptomic reference

Three aspects were considered while enhancing transcriptomic reference:

- **Gene overlaps** – some of the genes are overlapping, resulting in unassigned reads from those overlapping regions.

- **3' end extensions** – for certain genes, clusters of reads appear just beyond their annotated 3' ends. Pool et al. (2023) suggested that such reads may be biologically relevant and can be included by extending the 3' ends of affected genes.

- **Different annotations** – although most annotations agree on the majority of genes, a small fraction are present in some references but absent in others. To address this, multiple transcriptomic references were used during the enhancement process.

Overlaps The criteria for resolving overlapping genes were as follows:

1. **Gene type**: Protein-coding genes were preferred over other types, and long non-coding RNAs (lncRNAs) were preferred over the remaining categories (e.g., pseudogenes).

2. **Level**: Some references include a 'level' field indicating annotation quality: 1 for verified, 2 for manually annotated, and 3 for automatically annotated genes. Genes with lower scores were given higher priority.

3. **Intersection types**: If the 5' end of one gene overlapped with the 3' end of another, the 5' end gene was shortened.

Gene types were divided into three classes: protein-coding, long non-coding RNAs, and other. Each class was assigned a score, as shown in Table 5.3. The overall priority score was defined as the sum of the gene type score and the annotation level. A lower priority score indicates a higher-priority gene.

Intersection type (i.e., whether a 3' end is involved) was only considered in cases where the priority score were equal and the distance between the overlapping genes' 3' ends exceeded 1 kb. In such cases, lower-priority genes were either shortened or removed entirely, depending on whether shortening resolved the overlap (see Figure 5.4 for a graphical illustration).

Class	Score	Gene Types
protein coding	0	protein_coding, IG_C_gene, IG_D_gene, etc.
long non-coding RNA	10	lncRNA, lincRNA
other	20	snRNA, snoRNA, pseudogene, etc.

Table 5.3: Priority scores assigned to gene types for resolving overlaps.

3' end extensions Clusters of unassigned reads (with a size of at least 1 CPM) located within 100 bp downstream of a gene's 3' end were assumed to originate from that gene. Consequently, the corresponding gene annotations were extended to include these clusters.

Multiple annotations The initial transcriptomic reference (obtained from the 10x Genomics website) was incrementally enriched with gene entries from GENCODE, RefSeq (NCBI), and LNCipedia annotations. Specifically, entries from these sources were added if they overlapped with unassigned reads and did not overlap with genes already present in the current reference. The enrichment was performed sequentially: first incorporating relevant entries from GENCODE, followed by RefSeq, and finally LNCipedia.

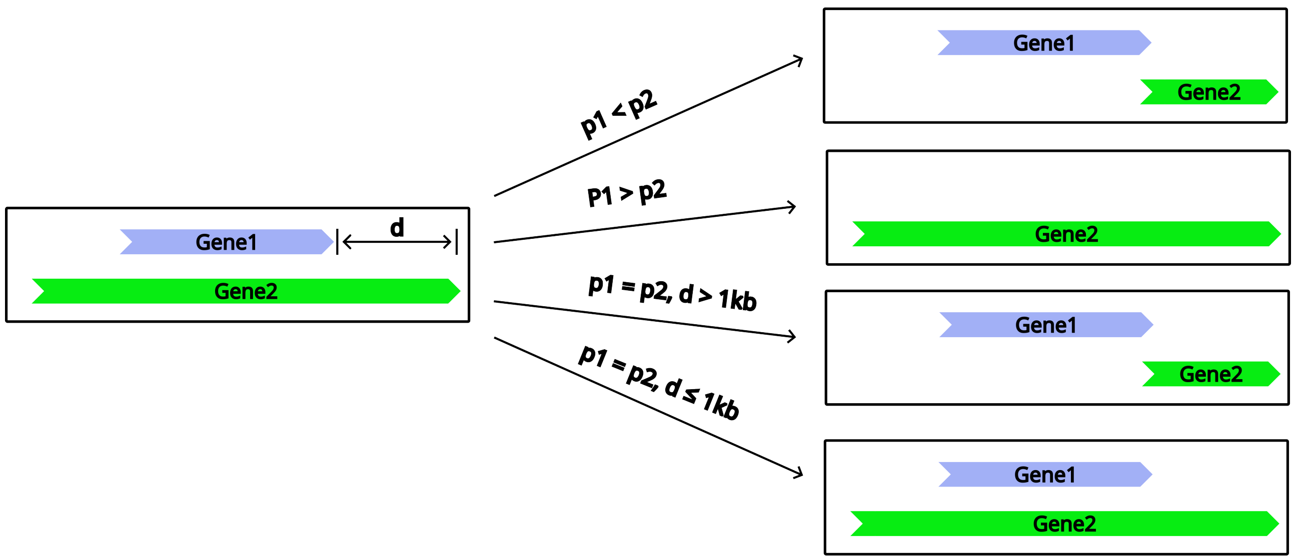


Figure 5.4: Strategy for resolving overlaps. *p1* and *p2* are priority scores of *gene1* and *gene2* (lower score means higher priority), while d is distance between 3' ends of those genes.

6. RESULTS

6.1 Intergenic regions

Fourteen datasets from various human tissues (blood, brain, eye, and lung) were analyzed. When using the standard transcriptomic reference (obtained from 10x Genomics website), between 11% and 29% of uniquely mapped reads remained unassigned (see Appendix 14.1). The majority of these unassigned reads were classified as intergenic, meaning they did not overlap with any annotated gene in the references used. Given the size of the datasets, this substantial proportion of intergenic reads suggests the potential for biologically relevant information to be uncovered.

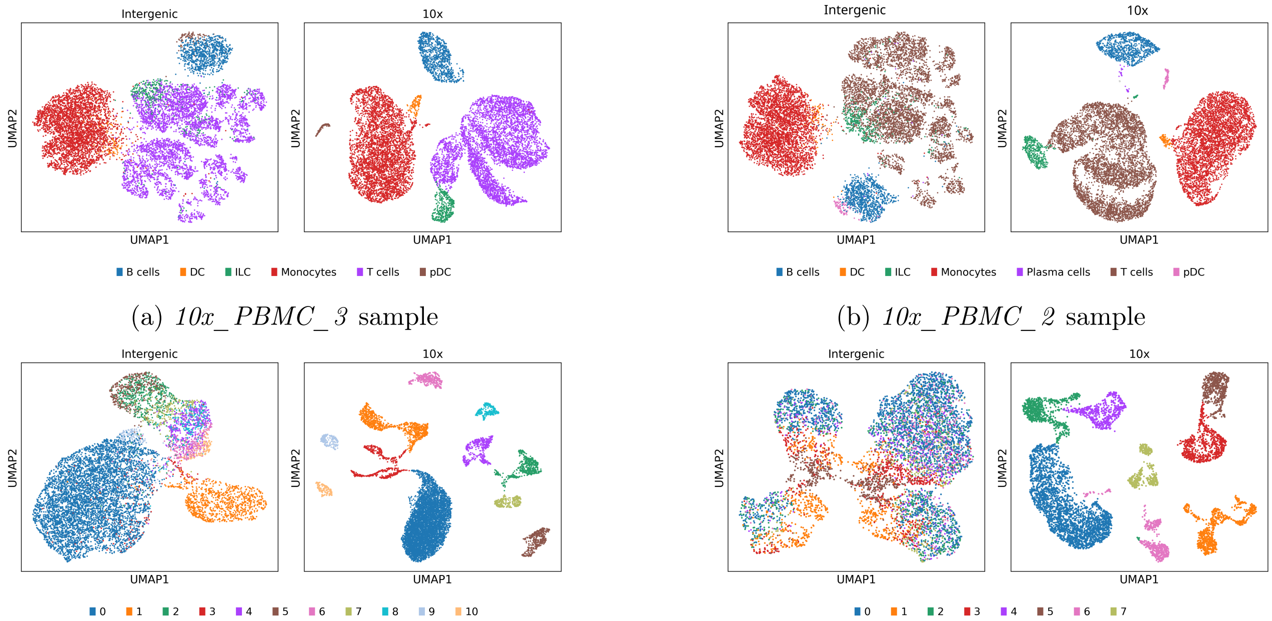
Analysing intergenic regions For each sample, intergenic regions containing a sufficient number of unassigned reads were identified as described in the Methods section, using a threshold of 5 CPM. The term *intergenic* is used in a strand-specific sense, i.e., regions located on the opposite strand of known genes are still considered intergenic.

To assess whether these regions carry biologically meaningful information, UMAP embeddings were generated for each sample using only reads originating from intergenic regions. As shown in selected examples (Figure 6.1), the resulting clustering patterns roughly corresponded to those obtained using the standard (10x Genomics) annotation. This suggests that intergenic reads capture underlying biological signal, or at least that the observed variation is structured and likely originates from real, cell-type-specific transcripts. Consequently, further analyses focused on these regions.

Intergenic regions from all samples were merged and filtered based on their recurrence across samples, resulting in a final set of 2,590 regions. Of these, 2,443 were located on the opposite strand of known genes and were classified as *antisense*, while the remaining 147 regions, which did not overlap any known gene on either strand, were classified as *isolated*.

poly-A sequences near the intergenic regions 3' end A poly-A region was defined as a stretch of at least 10 consecutive adenines, allowing one mismatch. If such a region is located near the 3' end of an intergenic region, it may indicate mispriming, suggesting that the observed read cluster does not represent the true 3' end of a novel transcript, but instead results from internal priming, and could therefore mark any internal position within the captured transcript.

This pattern was observed for the vast majority of identified intergenic regions: 99% contained a poly-A sequence within 200 bp of their 3' end. The lengths of these poly-A stretches ranged from 10 to 51 nucleotides with a mean length of 16.6 (see Figure 6.2).



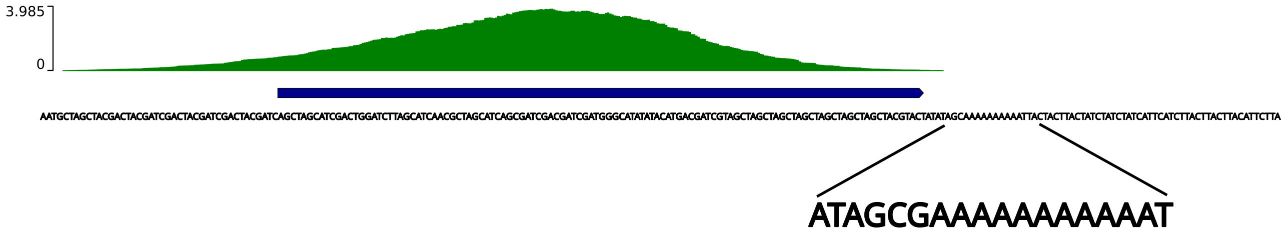
(a) *10x_PBMC_3* sample

(b) *10x_PBMC_2* sample

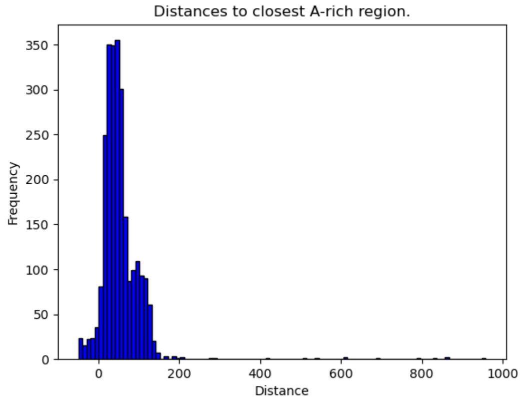
(c) *eye* sample

(d) *brain_2* sample

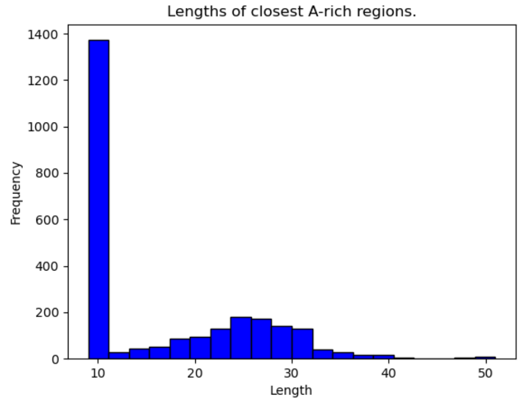
Figure 6.1: Comparison of clustering using the standard annotation (10x Genomics reference) and using only information from defined intergenic regions. In each plot, each dot represents a cell and is colored according to its cell type (cluster). Cell types (or clusters) were inferred using the standard annotation; the corresponding dots in the 'intergenic' plots were colored using the same labels. In some samples, visible clusters inferred using only intergenic regions roughly correspond to those obtained using the full standard annotation. In others (e.g., the *brain_2* sample), noise introduces clustering artifacts. Nevertheless, this rough clustering suggests that the unassigned reads do not represent purely random noise.



(a) Example of a poly-A segment located after the 3' end of a defined intergenic region.



(b) Distances from the 3' ends to the closest A-rich region downstream. Only 40 of the intergenic regions do not have such a region within 1 kb from the 3' end.



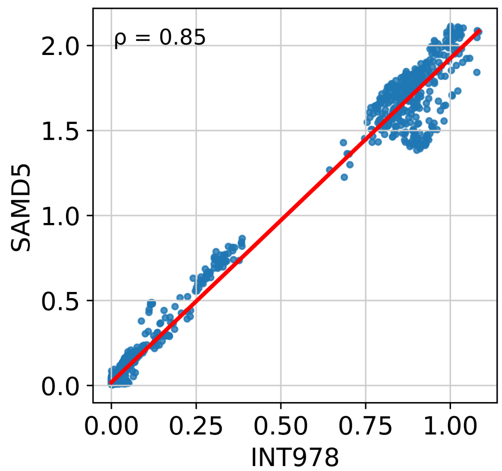
(c) Lengths of the closest poly-A segments downstream, measured in base pairs (bp).

Figure 6.2: Poly-A stretches (defined as at least 10 consecutive adenines, allowing one mismatch) are found near the 3' ends of the majority of defined intergenic regions.

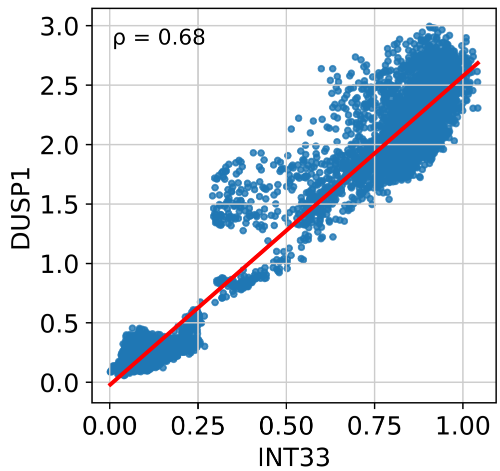
Correlations Another aspect to assess is whether the expression of these intergenic regions correlates with the expression of nearby genes. If such correlations exist, they may arise from several factors:

- If the region correlates with a gene on the same strand:
  - Technical issues – the genome annotation may be incomplete, possibly missing longer transcripts that span both the intergenic region and the gene with which the correlation is observed.
  - Biological reasons – genes in close proximity often exhibit similar expression profiles (Dong, Brown, and Truong 2024).
- If the region correlates with a gene on the oposite strand:
  - Technical artifacts – errors during library preparation could result in the production of inverted cDNA, which will be discussed later.
  - Biological reasons – genes located near each other tend to show similar expression profiles, potentially due to shared regulatory elements or similar chromatin structure (Zinani, Keseroğlu, and Özbudak 2022).

It is challenging to determine which of these scenarios applies to our samples. Of all the defined intergenic regions, 97 (3.74%) showed a Spearman correlation greater than 0.5 ($p < 0.05$) in at least one sample (83 correlated with genes on the opposite strand, 19 with genes on the same strand, and 5 with both). While these numbers are relatively small, it's important to note that scRNA-seq data is inherently noisy, and part of the correlations may be obscured by this noise. Some examples are provided in Figure 6.3.



(a) Correlation of *INT978* and *SAMD5* normalized expressions (Spearman correlation 0.85) in the *eye_3* sample, intergenic region is on the opposite strand of the gene.



(b) Correlation of *INT33* and *DUSP1* normalized expressions (Spearman correlation 0.68) in the *PBMC_10x_2* sample, both of them are on the same strand.

Figure 6.3: Examples showing correlations between intergenic regions and nearby genes.

6.1.1 Antisense intergenic regions

Interpreting antisense intergenic regions is complex because certain features typically associated with active transcription or gene sequences, such as open chromatin or conservation, are not strand-specific. These features could be attributed to the known genes present on the opposite strand of antisense regions. Therefore, only computational gene predictions were assessed as supporting evidence to distinguish antisense regions from potential artifacts.

Gene predictions To assess whether antisense intergenic regions overlap with predicted genes, UCSC gene prediction archives were examined, specifically focusing on overlaps with the 3' ends of predicted genes.

Out of the identified regions, 49 overlapped with predicted genes. Among them, 35 were located at the 3' end of predicted genes, which aligns with expectations for data derived from 3' end sequencing methods. Furthermore, 9 of these 35 predicted genes were longer versions of genes already annotated in the RefSeq or GENCODE references. The complete list can be found in Appendix 14.3.

RT artifacts Given that only a small fraction of antisense intergenic regions are supported by computational predictions, that the majority contain a poly-A stretch near the 3' end, and that some show strong correlation with genes on the opposite strand, we hypothesize that at least some of these regions may be artifacts introduced during library preparation.

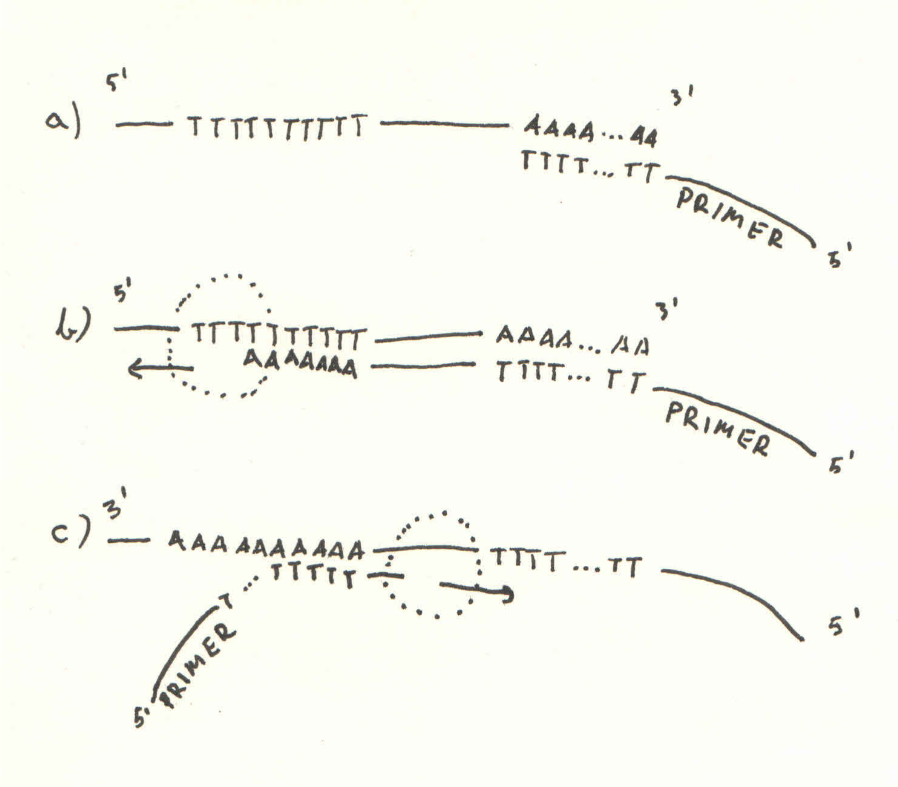


Figure 6.4: Schematic illustration of poly-A mispriming that can lead to reads mapping to the opposite strand of the original gene. a) An RNA molecule is captured via its poly-A tail by a primer containing a poly-T sequence. b) Reverse transcriptase binds the primer and synthesizes cDNA from the RNA template. c) If the RNA-cDNA hybrid dissociates, a second primer may bind to poly-A stretches within the newly synthesized cDNA. This can lead to synthesis of a reversed cDNA, ultimately causing the resulting reads to map to the antisense strand.

Specifically, poly-T primers may bind to internal poly-A stretches of cDNA synthesized from genuine transcripts, resulting in the creation of reverse-complement (inverted) cDNA fragments. These would then be mapped to the antisense strand of known genes (see Figure 6.4).

This hypothesis could also explain the clear clustering observed in Figure 6.1: while the unassigned reads appear to carry biological signal, it may actually reflect technical artifacts derived from real transcripts. Under this hypothesis, the unassigned data is not random noise but systematic artifacts derived from real, biologically relevant transcripts.

6.1.2 Isolated intergenic regions

In contrast, isolated intergenic regions cannot be explained by the mispriming hypothesis described above and may therefore represent novel genes. To assess whether these regions reflect genuine biological signals rather than technical artifacts, several aspects were evaluated:

1. Differential expression: are these regions differentially expressed in any samples, i.e., specific to certain cell types? Cell-type specificity would strongly support a biological origin.

2. Open chromatin: Are regions of open chromatin present upstream? Such regions are often indicative of active transcription initiation.

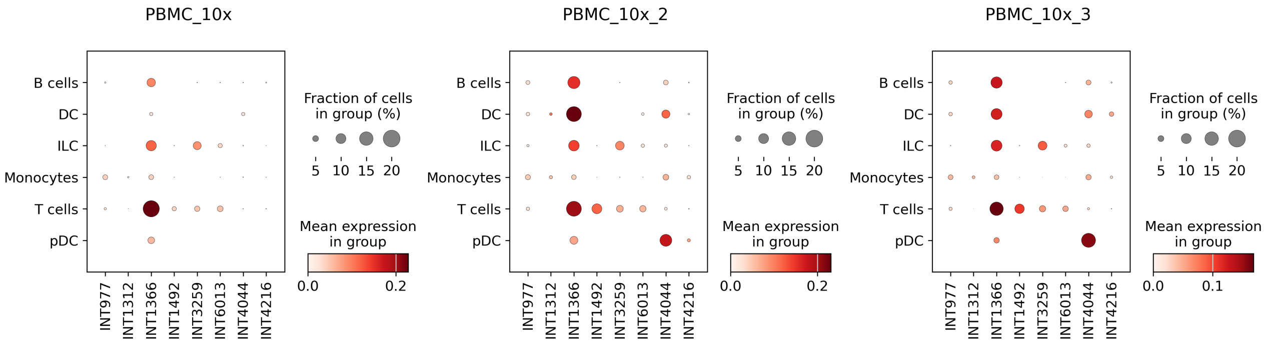


Figure 6.5: Examples of differentially expressed isolated intergenic regions in *PBMC_10x* samples.

3. Conservation score: Do these regions show evolutionary conservation? While coding regions are typically more conserved, non-coding functional elements can also display moderate conservation.

4. Gene predictions: Do these regions overlap with genes predicted by computational annotation tools?

Differential expression analysis To assess differential expression, cell clustering was performed for each dataset. For the *PBMC_10x* samples, clustering and cell type annotation were performed using *CellTypist*. For the remaining datasets, the Leiden algorithm was applied with manually selected parameters to yield a comparable number of clusters across similar samples. For example, all *lung* datasets were partitioned into five clusters, roughly aligning with observed UMAP structures.

Isolated intergenic regions were tested for differential expression using a threshold of 0.01 for adjusted p-value and 0.5 for log fold change. In total, 29 isolated regions passed these criteria. Examples are shown in Figure 6.5.

Open chromatin Open chromatin regions located upstream of intergenic regions may indicate transcriptional activity. However, their proximity to known genes must also be considered, as these open regions could be associated with known gene promoters rather than the intergenic regions.

For each isolated intergenic region, distances to the nearest open chromatin site and the nearest upstream gene (strand-agnostic) were computed. Among the 147 isolated regions, 66 had an open chromatin site closer than any known upstream gene. To further reduce the likelihood that these sites belonged to distant annotated genes, an additional filter was applied: the nearest open chromatin site had to be at least 2,000 bp away from the nearest known gene. This criterion was met by 40 intergenic regions.

Note that open chromatin information was derived from PBMC ATAC-seq data. Therefore, this feature should be interpreted cautiously for intergenic regions identified in non-blood tissue datasets.

Conservation scores Conservation scores reflect evolutionary conservation by comparing genomic sequences across multiple species. A high conservation score in an intergenic region may indicate functional importance and support the hypothesis of biological relevance.

To identify conserved regions, isolated intergenic regions with average conservation scores above 0.6 were classified as conserved. Only five such regions met this criterion. Overall, conservation

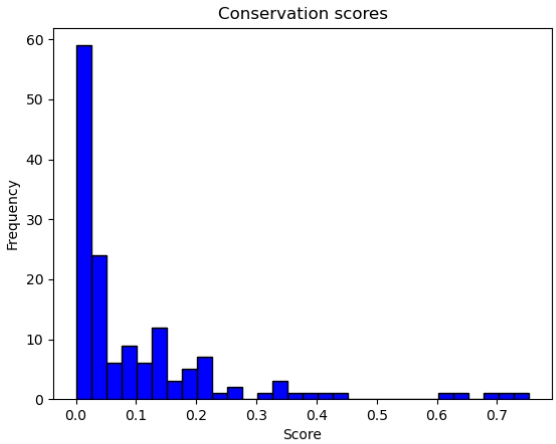


Figure 6.6: Distribution of average conservation scores across isolated intergenic regions.

scores across isolated regions were low (see Figure 6.6), suggesting that if these regions correspond to unannotated genes, they are likely to be human-specific.

Gene predictions To assess whether isolated intergenic regions overlap with predicted genes, UCSC gene prediction archives were examined. Twelve isolated regions overlapped with predicted genes. Of these, three did not align with the 3' ends of the predictions, which reduces the likelihood that they represent true transcript termini. Five predictions appeared to be extensions of known genes, while the remaining four matched genes not present in RefSeq or NCBI annotations but were supported by our data. A summary of these findings is provided in Appendix 14.2.

Combining supporting features Four features were examined as supporting evidence for the biological significance of isolated intergenic regions: conservation, open chromatin presence, differential expression, and gene predictions. Based on these features, four sublists of isolated intergenic regions were generated:

- Conserved regions
- Regions associated with open chromatin
- Differentially expressed regions
- Regions overlapping predicted genes

Notably, the list of conserved regions contained only one region (*INT1216*) that overlapped with the other lists, particularly the predicted gene list. The predicted gene list shared two regions (*INT196*, *INT1525*) with the differential expression list, and one region (*INT4147*) with the open chromatin list. All three of these predicted genes were extensions of already annotated genes (see example in Figure 6.7). The differential expression and open chromatin lists had 11 regions in common (*INT1312*, *INT1426*, *INT1492*, *INT1554*, *INT1609*, *INT1993*, *INT2209*, *INT349*, *INT368*, *INT4216*, *INT6013*). No regions appeared in more than two lists.

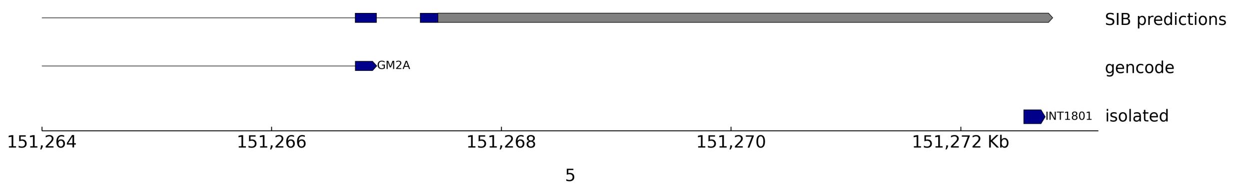
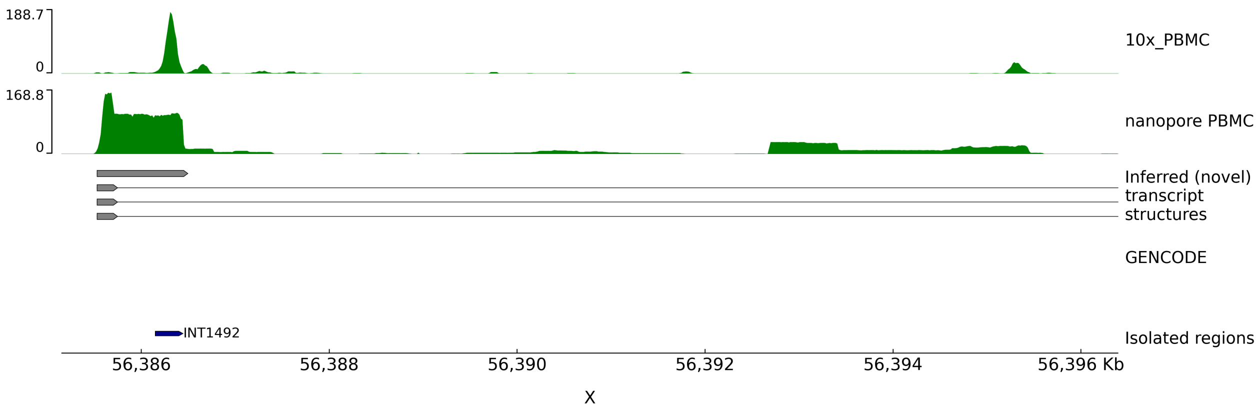
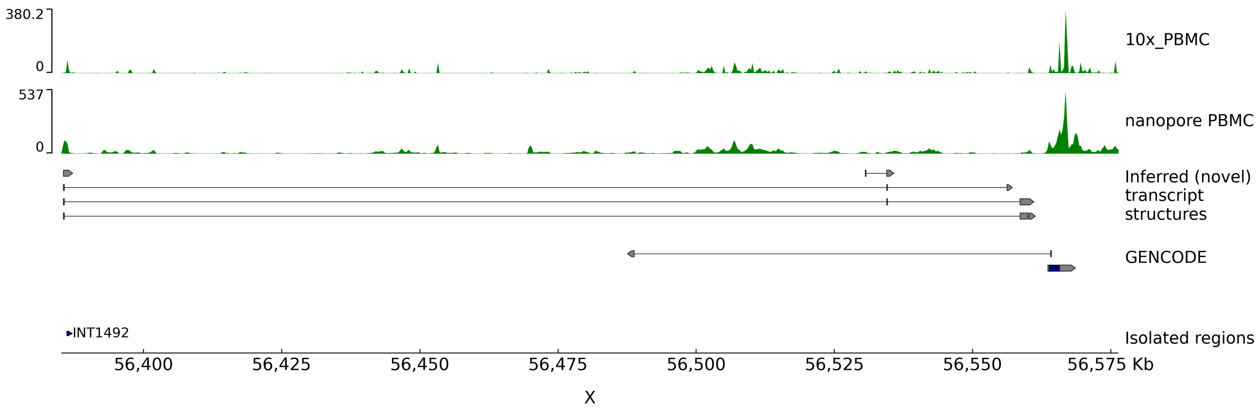


Figure 6.7: Example of a gene predicted to have longer transcripts than those currently included in the GENCODE and NCBI annotations. The intergenic region is located downstream of the 3' end of the *GM2A* gene. However, the SIB prediction tool suggests that the annotation of this gene should be extended to include longer transcript variants.

For comprehensive transcript structure inference, short-read data alone is insufficient. To investigate the feasibility of using long-read data, which generally has lower coverage and is less suitable for detecting rare transcripts, full-length Oxford Nanopore PBMC data were analyzed. This sequencing method successfully detected isolated transcripts, suggesting potential transcript structures. However, due to the low coverage, these inferred structures should be interpreted with caution. An example of the inferred transcript structures is shown in Figure 6.8.



(a) Inferred transcript structures for the *INT1492* region (zoomed-in view).



(b) Inferred transcript structures for the *INT1492* region (wider view).

Figure 6.8: Several transcript structures were inferred from the ONT data in the *INT1492* region, with transcript lengths ranging from approximately 1 kb to over 150 kb.

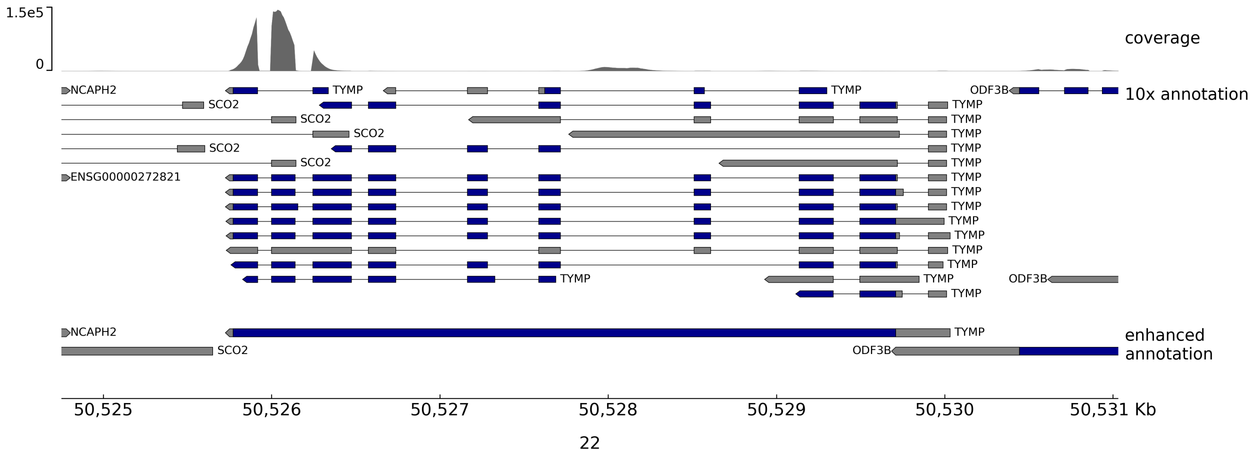


Figure 6.9: Example of successful resolution of overlapping genes, *TYMP* and *SCO2*. Both genes are protein-coding, but *TYMP* is preferred due to its lower *level* value. The exon structures clearly indicate that the reads in the overlapping region originate from the *TYMP* gene. Without reference enhancement, these reads would be classified as multimappers and filtered out in a typical workflow. The 10x annotation track is expanded to highlight the exon structures.

6.2 Enhancing transcriptomic reference

Given that unassigned reads can also map to gene locations, we sought to resolve overlapping genes in the annotations, allowing these reads to be included in downstream analysis. To address these overlaps, a straightforward set of rules was applied, prioritizing protein-coding genes over others and considering the "level" field in the transcriptomic annotations (more details are provided in Methods 5.5).

In addition to resolving overlaps, the original transcriptomic reference (from 10x Genomics) was supplemented with entries from other transcriptomic sources (RefSeq, GENCODE, LNCipedia), ensuring that new entries did not overlap existing ones. Furthermore, gene entries with reads near their 3' ends were extended.

This enhanced reference led to an average increase of 1.2% in the number of reads in the cell–gene matrices. Although the increase was modest, it enabled the detection of approximately 350 additional genes that were not identified using the standard transcriptomic reference (see Figure 6.9 for an example). The newly detected genes include transcription factors such as *SOX1*, *SOX2*, *YY2*, *HOXA10*, as well as other important genes like the growth factor *IGF2* and G-protein–coupled receptors (*GPR52*, *GPRASP1/2*). This suggests that adjusting the transcriptomic reference can help recover biologically important information.

However, on average, 13 genes per dataset were lost when using the enhanced reference. The combined list of lost genes contained 43 unique entries, the majority of which (35 out of 43) were lncRNAs, while 7 were protein-coding: *ABHD14B*, *KIF15*, *MICOS10*, *GSTM2*, *ATXN1L*, *ZIM2*, and *H2AC17*. This highlights that the resolution strategy is not perfect and should be applied cautiously.

Although the overall data volume increased, it is not straightforward to conclude that the data quality has improved. The overlap-handling rules themselves are open to debate, particularly regarding their applicability across all cases.

For example, genes may be expressed in certain cell types or states regardless of their level classifi-

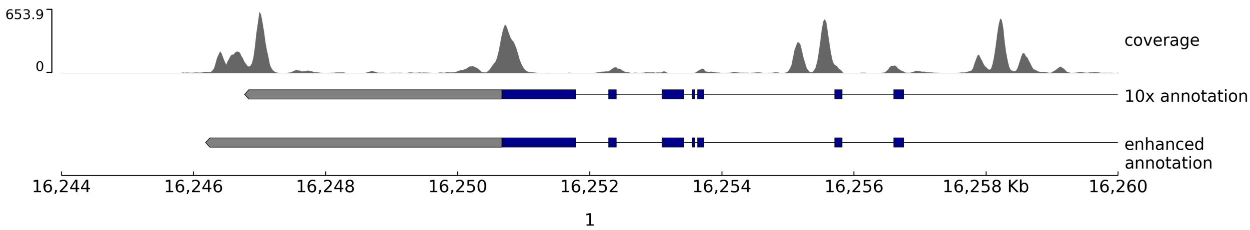


Figure 6.10: Example of gene extension (coverage shown is from the *PBMC_10x_2* dataset). In this case, the gene *FBX042* is extended to include reads near its 3' end. However, the number of reads gained is minimal compared to the total counts for this gene, with the read count increasing by approximately 3% in the *PBMC_10x_2* dataset.

cation (whether verified, manually annotated, or automatically annotated). While verified or manually annotated genes are generally expressed more frequently than automatically annotated ones (which are likely less explored and detected at lower rates), this pattern does not hold in every case. The same consideration applies to gene classification by type.

Regarding the gene extensions, in most cases, they only lead to a modest increase in gene expression counts (see Figure 6.10 for an example). While some additional information is gained, the increase is relatively small compared to the total data volume. For instance, in the *PBMC_10x_2* sample, the gain from gene extensions constituted only 1.76% of the additional reads or 0.017% of the total reads.

Overall, while statistical differences were observed, their impact on biological interpretation remains uncertain and likely depends on the specific downstream analysis. For example, in the *PBMC_10x_2* sample, 99.9% of cells were assigned the same cell types using both the standard and enhanced references. However, in some cases, even a small number of additional genes could significantly influence the analysis, suggesting that the enhanced reference might help uncover relevant biological processes. For instance, the recovered gene *SOX1* plays a direct role in neural cell fate determination and differentiation (Kan et al. 2004), and may therefore be highly informative in developmental studies.

7. DISCUSSION

Single-cell RNA sequencing has become a widely used tool for studying cellular systems. These experiments generate large volumes of data, and as the technology becomes increasingly affordable and accessible, the total amount of data produced is expected to grow rapidly. While scRNA-seq data is primarily used to quantify annotated transcripts, a substantial fraction of reads that map to the genome but do not overlap known features are typically excluded from analysis. As demonstrated in the Results chapter, such unassigned reads can account for 10–30% of the total reads, representing a significant portion of the dataset.

This project was designed to explore these unassigned reads. They were first categorized based on whether they overlapped known genes. Those that did not were termed intergenic. Intergenic reads were further divided into two categories: antisense (mapped to the opposite strand of known genes) and isolated (not overlapping any known genes on either strand). Reads were clustered, and only regions with a minimum expression threshold of 5 CPM were considered for analysis.

Antisense intergenic regions Two key observations support the hypothesis that these reads are artifacts generated during reverse transcription: (i) their expression levels are correlated with those of genes located on the opposite strand, and (ii) poly-A stretches are frequently observed downstream of these regions. As these reads constitute the vast majority of unassigned reads, understanding this mechanism could improve the efficiency of single-cell RNA-seq experiments. This hypothesis is particularly relevant to scRNA-seq protocols relying on poly-T primers.

Temperature modifications might help mitigate such artifacts: RNA–cDNA complexes are more stable at lower temperatures. In the 10x Genomics v3.1 protocol, reverse transcription occurs at $52°C$, while in the inDrops2 protocol it is performed at $42°C$, where fewer unassigned reads were observed (see Appendix 14.1). However, strong conclusions cannot be drawn, as only two inDrops2 samples were analyzed.

While this hypothesis may account for most antisense reads, other sources of noise should not be excluded. For example, reverse transcription may introduce artifacts through template switching (Balazs et al. 2019) or sense–antisense transcript fusion (Houseley and Tollervey 2010). Additional errors can also arise during the amplification (PCR) step (Potapov and Ong 2017).

Isolated intergenic regions Unlike antisense regions, isolated regions may represent unannotated (novel) genes rather than artifacts. This is particularly likely for regions with supporting evidence, with the most indicative feature being differential expression — as noise is not expected to be cell-type specific.

For example, the *INT1492* region, located 70 kilobases from the nearest known gene, exhibited

T-cell specificity and showed upstream open chromatin, strongly suggesting it may represent a real T cell–specific transcript. Another example, *INT1609*, was differentially expressed in lung datasets and also supported by open chromatin accessibility. Some regions overlapped with the 3' ends of genes predicted by the SIB tool (e.g., *INT827*, *INT4070*, *INT4577*, and *INT4948*), while others showed moderate conservation scores (e.g., *INT1829*, *INT2199*, *INT3636*).

Although the human genome has been extensively studied, several factors continue to complicate the identification of novel genes, especially long non-coding RNAs (lncRNAs). lncRNAs often lack features that enable reliable computational annotation. For example, they are typically not conserved, lack long open reading frames (unlike protein-coding genes), and frequently exhibit non-canonical splicing. In addition, their low expression levels and high cell-type specificity hinder confident annotation — a requirement for inclusion in widely used transcriptomic references such as GENCODE or RefSeq.

While transcript structures can be inferred using long-read RNA sequencing (as demonstrated with Oxford Nanopore data), high sequencing depth is necessary for robust annotation.

Although lncRNAs can participate in virtually any cellular process, predicting the function of a specific lncRNA remains challenging. They often lack clear phenotypic effects, and mutations usually affect quantitative traits (Mattick et al. 2023). This project did not attempt to assign functions to the novel candidates; future studies could explore this further.

A subset of isolated regions located near known genes may instead represent unannotated transcript extensions. Indeed, several regions (e.g., *INT196*, *INT1525*, *INT1801*, *INT2044*, and *INT4147*) overlapped with predicted longer isoforms of known genes, supporting this as a possible explanation. Long-read sequencing data could help validate this hypothesis.

Enhancing transcriptomic reference Some reads remained unassigned despite aligning to annotated gene regions, primarily due to overlapping genes. To incorporate such reads, we enhanced the transcriptomic reference by resolving overlaps based on gene types. Using this enhanced reference, the downstream cell–gene matrices increased by approximately 1.2% on average. Notably, several biologically important genes, such as transcription factors, were recovered using this approach.

While the overall data volume increased (in both the number of detected genes and total counts in cell–gene matrices), some genes showed reduced counts after overlap resolution — highlighting the complexity of human genome and the difficulty of disambiguating overlapping genes using short reads.

This method of enhancing transcriptomic references raises several points of consideration. First, prioritizing certain gene types (e.g., protein-coding over lncRNAs) can be subjective and may not always be appropriate. Second, our overlap resolution was based on gene-level annotations (start and end coordinates), whereas transcriptomic annotations typically include more detailed exon–intron structures and isoforms.

Alternative strategies for handling multi-mapped reads exist, such as uniform distribution or probabilistic assignment based on the ratio of uniquely mapped reads (Deschamps-Francoeur, Simoneau, and Scott 2020). However, no consensus or gold standard currently exists, and such reads are often excluded to reduce uncertainty.

A more refined approach could incorporate transcript-level annotations, e.g., comparing read coverage with exon structures to infer the most likely gene of origin in overlapping regions. Such a tool could recover more reads and improve assignment accuracy.

Limitations and discrepancies This project analyzed datasets generated using short-read, 3'-end capturing RNA-seq methods. Datasets produced by other methods may contain different artifacts and would require separate analysis. Additionally, the data used here do not support full transcript structure reconstruction, which is possible with full-length sequencing.

An interesting discrepancy is shown in Table 14.1: the proportion of unassigned reads is lower when calculated from raw counts compared to demultiplexed counts. Since demultiplexed counts reflect actual molecule numbers and raw counts reflect amplification, one would expect similar ratios — assuming equal amplification efficiency across read types. However, we consistently observed a higher proportion of unassigned reads in the demultiplexed data, suggesting that these reads may be amplified less efficiently. The source of this bias remains unclear and warrants further investigation. This effect is especially pronounced in datasets generated using the inDrops2 protocol, so comparisons between inDrops2 and 10x Genomics v3.1 may help identify its cause.

8. CONCLUSIONS

- The analyzed scRNA-seq datasets contained unassigned reads from both gene and intergenic regions. The majority of these reads (68% on average) originated from intergenic regions.
- The intergenic regions were classified into two groups:
  - Regions located on the opposite strand of known genes. Reads from these regions constitute the majority of unassigned reads and are likely artifacts generated during the reverse transcription step.
  - Regions without annotated genes on either strand. Reads from such regions may originate from unannotated genes. This thesis proposes 147 candidate regions, 55 of which are supported by at least one additional line of evidence.
- Some unassigned reads that mapped to regions with overlapping genes may be recovered by resolving gene overlaps in the transcriptomic annotation. The proposed approach increased the number of counts in the cell–gene matrices by approximately 1.2%.

9. DISCLAIMER

Parts of this thesis were reviewed using OpenAI's ChatGPT to assist with grammar and style corrections. The scientific content, interpretations, and conclusions are solely my own.

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12. SUMMARY

Vilnius University
Systems Biology

ENHANCING SINGLE-CELL RNA-SEQ ANALYSIS THROUGH THE INTEGRATION OF UNACCOUNTED INTERGENIC REGIONS

Juozapas Ivanauskas

Abstract

Single-cell RNA sequencing (scRNA-seq) experiments generate large datasets, a portion of which often consists of reads that align to the genome but are not assigned to any annotated gene. This project aimed to investigate the origins of these unassigned reads and explore strategies to recover them for downstream analysis.

Unassigned reads from fourteen publicly available scRNA-seq datasets were analyzed. The majority were found to originate from the antisense strand of known genes. It is hypothesized that these reads are artifacts introduced during the reverse transcription step, particularly in protocols using poly-T primers. Another subset of reads mapped to intergenic regions without any annotated genes on either strand. These regions may correspond to unannotated transcripts, and 147 such candidate regions were identified for future validation.

Additionally, a fraction of unassigned reads mapped within known gene regions but were not assigned due to overlapping annotations. To recover these, the transcriptomic reference was enhanced by resolving gene overlaps, extending 3' ends of selected genes, and combining information from four different annotations. This modification led to an average increase of 1.2% in total read counts in the resulting cell–gene matrices.

Together, these findings suggest that unassigned reads—often discarded in scRNA-seq workflows — can be informative and should be systematically examined to improve transcriptomic references and the interpretability of single-cell data.

13. SUMMARY IN LITHUANIAN

Vilniaus Universitetas
Sistemų Biologija

PAVIENIŲ LĄSTELIŲ RNR SEKOSKAITOS DUOMENŲ ANALIZĖS TOBULINIMAS INTEGRUOJANT NEAPRAŠYTUS TARPGENINIUS REGIONUS

Juozapas Ivanauskas

Santrauka

Pavienių ląstelių RNR sekoskaitos eksperimentai sugeneruoja didelius duomenų kiekius, kurių dalį sudaro sekos, prilygiuojamos prie genomo, bet nepriskiriaamos jokiam genui. Šio projekto tikslas buvo ištirti šias sekas ir pabandyti įtraukti jas į tolimesnes analizes.

Šiame projekte buvo išanalizuota nepriskirtos sekos iš keturiolikos duomenų rinkinių. Dauguma jų buvo priligiuojamos prie priešingos DNR grandies nei žinomi genai. Buvo iškelta hipotezė, kad šios sekos yra artefaktai susidarantys atvirkštinės transkripcijos metu (ši hipotezė galioja tik sekoskaitos metodams naudojantiems poli-T pradmenis). Kita dalis sekų buvo priligiuota prie genomo vietose, kuriose iš viso nėra žinomų genų. Šios sekos galimai kyla iš neaprašytų genų. 147 regionai buvo iškelti kaip kandidatai naujiems genams, kuriuos reikėtų patvirtinti papildomais ekperimentais.

Galiausiai, dalis sekų buvo priligiuojamos tose vietose, kur yra žinomų genų, tačiau vis tiek liko nepriskirtos. Kad įtrauktume šias sekas į tolimesnę analizę, patobulinome transkriptominį anotacijos failą. Tai buvo pasiekta išsprendžiant persidengiančius genus, pratęsiant kai kurių genų 3' galus bei naudojant informaciją iš kelių skirtingų transkriptomo anotacijų. Taikant šį metodą, duomenų kiekis tolesnėje analizėje padidėjo vidutiniškai 1.2%.

Šie rezultatai rodo, kad nepriskiriamos sekos (įprastai neįtraukiamos į tolimesnes analizes), gali būti informatyvios ir turėtų būti sistemiškai išanalizuotos siekiant pagerinti dabartines genomo anotacijas bei pavienių ląstelių duomenų analizę.

14. APPENDICES

Sample	Total Reads	Unassigned	Intergenic	Demultiplexed	Unassigned	Intergenic
PBMC_10x	182330834	15.95%	11.37%	70305137	18.51%	12.86%
PBMC_10x_2	496387931	18.52%	14.26%	195948926	21.29%	16.32%
PBMC_10x_3	368640939	18.59%	14.34%	167804193	20.99%	16.17%
PBMC_indrops	112932507	11.12%	6.5%	8176578	22.86%	10.25%
PBMC_indrops_2	471705924	14.87%	8.36%	98063027	25.52%	13.73%
brain	206360627	16.87%	10.67%	143916177	19.02%	12.07%
brain_2	122556503	22.32%	15.98%	95031864	23.77%	17.00%
eye	375397270	28.02%	18.97%	161882445	31.87%	21.60%
eye_2	140981808	29.35%	14.85%	69146995	31.43%	15.99%
eye_3	161261977	29.46%	18.86%	68157696	32.11%	20.58%
lung_2	511080104	15.99%	12.07%	269118747	17.98%	13.33%
lung_5	452105505	14.49%	10.74%	256353198	16.19%	11.83%
lung_7	524095146	18.41%	14.34%	236157103	20.64%	15.79%
lung_8	342092138	14.53%	10.66%	217801055	16.07%	11.67%

Table 14.1: Statistics of unassigned and intergenic reads per sample. Unassigned reads were counted after mapping with 10x reference. Intergenic reads here are those that do not intersect with any reference used. 'Demultiplexed' column shows number of total reads after demultiplexing using UMIs, and following 'unassigned' and 'intergenic' percentages were computed compared to this number.

Name	Genomic coordinates	Prediction tool	Overlaps with 3' region of predicted gene
INT196	6:89086200-89086500	SIB	Yes (PNRC1)
INT827	9:94111950-94112200	SIB	Yes
INT1216	1:72282700-72282900	SIB	No (5' end)
INT1387	11:63570800-63571050	SIB	No
INT1525	8:11842200-11842350	SIB	Yes (CTSB)
INT1801	5:151272550-151272700	SIB	Yes (GM2A)
INT2044	19:4041150-4041400	SIB	Yes (ZBTB7A)
INT4070	4:47430500-47430700	SIB	Yes
INT4147	1:34861500-34861700	SIB	Yes (DLGAP3)
INT4577	5:702050-702300	SIB	Yes
INT4948	5:703250-703450	SIB	Yes
INT5710	4:47428500-47428700	SIB	No

Table 14.2: Isolated intergenic regions overlapping with predicted genes from UCSC gene prediction archive. The gene in the brackets shows if the predicted gene is extended version of already annotated genes.

Table 14.3: Antisense intergenic regions overlapping with predicted genes from UCSC gene prediction archive. The gene in the brackets shows if the predicted gene is extended version of already annotated genes.

Name	Genomic coordinates	Prediction tool	Overlaps with 3' region of predicted gene
INT7	6:24718850-24719100	SIB	Yes
INT33	5:172767250-172767500	SIB	Yes (DUSP1)
INT134	15:41280550-41280800	SIB	Yes (extends a bit after)
INT218	5:61409150-61409400	SPG	No (5' end)
INT285	10:110898200-110898550	SIB	Yes (BBIP1)
INT327	1:169132100-169132350	SIB	Yes (NME7)
INT347	14:75277700-75278000	Gescan	No
INT382	12:11892250-11892500	Gescan	Yes
INT386	6:26215200-26215450	SIB	Yes (H2BCB?)
INT438	4:39713300-39713600	SIB	Yes (almost all (short) gene spanned)
INT533	8:130110900-130111150	SIB	Yes
INT617	15:85733950-85734200	SIB	Yes
INT621	9:75148850-75149400	SIB	Yes (OSTF1)
INT651	1:28579100-28579350	SIB	Yes (TRNAU1AP)
INT654	6:34382850-34383050	SIB	No (but short gene)
INT828	8:133035800-133036050	SIB	Yes (SLA)
INT859	2:71373500-71373750	SIB	No (slight overlap on 5' end)
INT898	17:51190400-51190700	SIB	Yes (extends after)
INT903	8:22613400-22613650	SIB	Yes
INT949	9:111693050-111693400	SIB	Yes
INT972	15:64954800-64955350	SIB	Yes
INT984	18:63291900-63292200	Geneid	Yes
INT1028	5:50414200-50414500	SIB	Yes
INT1155	1:36296050-36296300	SIB	No (5' end)
INT1231	10:19890700-19891050	Gescan	Yes
INT1402	5:142786700-142786950	SIB	No (5' end)
INT1440	2:235746450-235746700	SIB	Yes
INT1445	9:40885650-40885900	SIB	Yes
INT1548	7:50342700-50342950	Gescan	No (exon in the middle)
INT1616	17:30895450-30895700	SIB	No (5' end)
INT1749	3:172333550-172333750	SIB	No (5' end)
INT1931	17:82522100-82522350	SIB	Yes
INT2065	10:103606150-103606400	SIB	Yes
INT2158	14:49636100-49636350	SIB	No (5' end, DNAAF2)
INT2194	7:130521250-130521500	SIB	No (but gene is short)
INT2371	1:14632650-14632900	Geneid	No (middle exon)
INT2380	22:33853900-33854150	Geneid	Yes
INT2825	13:45300050-45300300	Geneid	No (middle exon)
INT2979	4:98929350-98929600	SIB	No (5' end, EIF4E)
INT3328	2:184604250-184604500	SIB	Yes
INT3429	17:48172750-48173150	Augustus	Yes
INT3504	15:25332950-25333150	SIB	Yes (UBE3A)
INT3849	3:121432200-121432400	SIB	Yes
INT3868	17:31448500-31448700	SIB	Yes
INT4458	5:44820700-44820900	SIB	Yes (MRP530)
INT4743	20:3188900-3189050	SIB	Yes (DDRGK1)
INT5002	11:92232750-92232950	SIB	Yes
INT5315	20:3183750-3184050	SIB	Yes
INT5329	13:95603000-95603200	SIB	Yes (extends after)